Multi-Text Representation Learning for Patent Validity Prediction

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October 2026

Abstract

Patents can account for a substantial share of firm value, making the prediction of patent validity litigation outcomes economically important. Predicting these outcomes requires interpreting interdependent texts and aggregating information across multiple validity challenges. Using a newly constructed dataset of UK patent litigation, I develop a multiple-instance learning architecture tailored to the structure of these challenges. The architecture uses cross-attention, constructing queries from representations of the challenged patent claim and keys and values from representations of the texts cited against it. To account for textual dependencies, I construct expanded text representations that explicitly incorporate relevant content from the texts on which they depend. Together, these components address how individual texts are interpreted and how information across challenges is combined into an overall validity prediction. Empirical results show that the proposed architecture outperforms benchmark architectures from the literature as well as ChatGPT in this setting. I also propose a novel quantitative method for evaluating the faithfulness of attention-based explanations, i.e., the extent to which they reflect the model's actual prediction process.

1 Introduction

Patents can be economically important intangible assets. Empirical evidence links patents to firms' market value (Hall et al., 2005) and shows that patent litigation outcomes can have substantial financial consequences (Panattoni, 2011). Predicting these outcomes can therefore help assess the legal uncertainty surrounding patent rights. Since validity depends partly on patent content, this paper uses natural language processing to predict claim-level patent validity in U.K. litigation.

My dataset contains U.K. patent litigation judgments from 1995 through August 2026, covering 405 cases and 561 patent invalidity events. My first contribution is to develop LLM prompts for extracting structured information from these judgments. A patent's description explains the invention, while its claims define the scope of protection. In litigation, validity is assessed at the claim level; my predictions are conditional on the attacks brought against each claim.

This task raises a broader modeling question: How should a model represent text whose meaning depends on other text? A dependent claim inherits the requirements of the claim it references and adds further restrictions. For example, if claim 1 requires features A and B, claim 2 may simply

refer to claim 1 and add feature C. Claim 2 therefore requires all three features, even though its own text does not explicitly repeat A and B.

Much of the literature either focuses on independent claims or treats dependent claims as self-contained, although Gao et al. (2024) and PEDANTIC (Knappich et al., 2025) explicitly address dependencies. I introduce *expanded claims*, which make inherited limitations explicit so that each dependent claim can be read independently.

Predicting a claim's validity also requires combining evidence across the attacks brought against it. I formulate this task as a multiple-instance learning (MIL) problem: each bag contains the attacks against a claim, and its label indicates whether that claim is invalid. I develop an architecture that encodes the challenged claim separately from the texts associated with each attack. These representations interact through cross-attention, with queries derived from the claim representation and keys and values derived from the attack-specific representations. This design allows the model to weight and combine evidence according to its relevance to the challenged claim.

Finally, I propose a novel regression-based method to measure how attention weights affect model predictions. Previous studies have examined this relationship through attention perturbation and erasure (Jain and Wallace, 2019; Serrano and Smith, 2019; Wiegreffe and Pinter, 2019; DeYoung et al., 2020). However, these intervention results are often presented as figures and summary statistics without corresponding statistical tests. This makes it difficult to assess whether the reported differences are statistically significant. Applying my method to my dataset, I find that larger attention reallocations are associated with larger prediction changes, and that this relationship is statistically significant. These results provide statistical evidence that attention allocation affects predictions in my proposed architecture.

One of the biggest distinctions between my prediction setting and those used in studies such as PANORAMA (Lim et al., 2025) and FiNE-Patents (Knappich et al., 2026) is the information set being used. My approach assumes a worst-case scenario for prediction, in which no additional information about the judge's analysis is used. This distinction is important for my goals: observers do not have access to the judge's analysis until the judgment is issued.

The remainder of the paper is organized as follows. The remainder of Section 1 reviews the related literature. Section 2 describes the U.K. patent system and the institutional and legal setting. Section 3 explains the construction of the dataset, the outcome definitions, and the training and evaluation samples. Section 4 describes the architecture and evaluation metrics. Section 5 presents claim-level prediction results for the court and UKIPO holdouts. Section 6 examines how attention weights affect the model's predictions. Section 7 concludes.

1.1 Related Literature

1.1.1 Machine-learning literature

Prediction settings. Relatively few studies directly predict outcomes in patent litigation. Much of the related literature instead examines patent-office decisions on whether to grant a patent or reject an application on a particular ground (Lo and Chu, 2021; Ashtor, 2022; Knappich et al., 2025;

Lim et al., 2025; Knappich et al., 2026). Studies of litigation outcomes include Janicke and Ren (2006), Cowart et al. (2014), and Hicks (2021). These earlier studies employ logistic regression, trees, or TF–IDF, without evaluating transformer-based language models. To my knowledge, my paper is the first study to evaluate transformer-based language models for predicting patent-validity outcomes in a litigation setting.

Information available for prediction. A second distinction concerns the information available when predictions are made. Raghupathi et al. (2018) and Raghupathi (2026) apply text-mining and classification methods to published judicial opinions in pharmaceutical patent cases. These texts contain the court's reasoning, which is unavailable before judgment. A related distinction arises in patent examination: PANORAMA's patentability-classification task supplies passages from prior-art documents that the examiner cited in the office action (Lim et al., 2025). Although valuable in their own right, these exercises do not directly address the problem faced by traders and lawyers seeking to assess patent vulnerability before the relevant decision is issued.

Hicks (2021) provides the closest comparison to my study in terms of the prediction task, predicting judicial validity outcomes from claim text without relying on the court's written reasoning. He studies U.S. district-court cases filed in 2008 and 2009, considering specifications based on either the complete claim set or the asserted claims. He examines validity outcomes at both the patent and ground-of-attack levels, including novelty, obviousness, and indefiniteness. However, his textual inputs do not include the patent description or cited prior-art documents. My study extends the range of texts examined and evaluates their predictive value using alternative representations and models. Published judgments supply outcome labels and identify the claims, grounds pursued, and documents relied upon at the hearing, while judicial reasoning and statements of the outcome are excluded from the predictive inputs.

Representing dependent claims. Claim dependencies create an additional challenge because the literal text of a dependent claim does not state all the limitations defining its scope. Gao et al. (2024) explicitly model these dependencies when predicting novelty rejections from USPTO prosecution data. They decompose claims into textual nodes and connect them through a graph representing relationships within claims and across parent and ancestor claims. Fixed text embeddings initialize the nodes, and graph neural networks combine information across connected nodes to produce a prediction. PEDANTIC (Knappich et al., 2025) addresses the problem using a different approach. They use a retrieval-augmented agent that can obtain relevant passages from parent claims when interpreting the target claim.

Multiple-instance learning. MIL provides a framework for predicting outcomes for bags of instances (Dietterich et al., 1997; Maron and Lozano-Pérez, 1998). A prominent application is whole-slide image classification, where information from image patches is aggregated into a slide-level prediction. Ilse et al. (2018) develop attention-based pooling of instance representations, while TransMIL incorporates morphological and spatial correlations among patches (Shao et al., 2021).

In my setting, each bag represents a claim and contains the attacks brought against it. Every instance within the bag contains exactly the same challenged claim and full claim set, while the grounds of challenge and associated texts may vary. This sharing of entire input components is distinct from correlations among image patches and motivates an explicit separation between information common to the bag and information specific to each attack. Similar settings appear in Sathe and Park (2021) and Kumar et al. (2023). The closest architectural connection is Xu et al. (2025), who use cross-attention with the query as Q and the bag instances as K and V. My architecture follows a similar principle, using the challenged claim as the query and the attack-specific representations as keys and values.

1.1.2 Economics literature

The economics literature provides two motivations for predicting patent-validity outcomes: patents can generate substantial value for their owners, and the loss of protection can affect competition and subsequent innovation. These studies establish the economic consequences of patent protection; my paper examines whether patent texts and related information can predict whether challenged claims survive.

The first branch examines the value of patents to their owners. Pakes (1986) estimates annual returns from patent protection at 11–16% of R&D expenditure, with half accruing to 5–10% of patents. Hall et al. (2005) find that, conditional on R&D and patent stocks, one additional citation per patent is associated with approximately 3% higher firm value at the sample mean. Kogan et al. (2017) find evidence of creative destruction: high-value patents benefit their owners, while capital and labor move from non-innovating firms toward innovating firms. Farre-Mensa et al. (2020) show that patent grants accelerate startup growth and improve access to external financing.

The second branch studies the economic consequences of losing patent protection. Marco (2005) estimates that invalidity decisions reduce owners' stock-market value by approximately 1.4%. Panattoni (2011) finds cumulative abnormal market-value losses of slightly more than $1 billion for branded pharmaceutical firms losing patent litigation in the sample. Following patent expiration and generic entry, Suh et al. (2000) find that originators' dollar sales fall by approximately 12% by the fourth year and their revenue market share falls to 78% of its initial level. Branstetter et al. (2016) show that generic entry following successful patent challenges reduces prices and incumbents' market shares while increasing competition and consumer surplus. Galasso and Schankerman (2015) find that invalidation increases follow-on innovation, measured by subsequent citations, in computers, electronics, and medical instruments, with no significant effect in drugs, chemicals, or mechanical technologies.

2 Overview of the UK Patent System

2.1 Institutions and procedural settings

In England and Wales, first-instance patent litigation is heard principally by the Patents Court and the Intellectual Property Enterprise Court (IPEC). The UKIPO Comptroller also determines certain patent disputes as an administrative tribunal, with most decisions appealable to the Patents Court. Patents Court and IPEC judgments may be appealed, with permission, to the Court of Appeal and potentially onward to the UK Supreme Court.[1]

2.2 The specification and the claim-level nature of validity

A patent specification contains a description of the invention, claims, and any relevant drawings. The description explains the invention and how it may be performed, while the claims define the scope of protection. An independent claim stands on its own. A dependent claim incorporates the limitations of an earlier claim and adds further limitations. Validity is assessed claim by claim, so some claims may survive while others are invalidated.

Throughout this paper, I use *Leo Pharma v Sandoz* as a running example.[2] The dispute concerned European Patent (UK) EP0679154B1, covering calcipotriol monohydrate and pharmaceutical compositions containing it. Example 2.1 presents its five claims in abbreviated form.

Example 2.1 (Independent and dependent claims)**.**

1. Calcipotriol monohydrate.
2. A pharmaceutical composition containing the compound of claim 1.
3. A pharmaceutical composition according to claim 2 that is a cream.
4. A pharmaceutical composition according to claim 2 that is a gel.
5. A pharmaceutical composition according to any one of claims 2–4, with an active-component content of 1–100 μg/g of the composition.

Claim 1 independently defines the compound. Claim 2 incorporates claim 1 and adds the requirement of a pharmaceutical composition. Claims 3 and 4 restrict claim 2 to particular dosage forms, while claim 5 adds a concentration restriction to any one of claims 2–4.

Invalidating claim 1 does not automatically invalidate claims 2–5. Each challenged claim must therefore be assessed on its own limitations and against the grounds raised.

[1] See UK Intellectual Property Office, nd, secs. 18, 72, 74, and 97, Courts and Tribunals Judiciary, ndb, Courts and Tribunals Judiciary, nda, and UK Supreme Court, nd.

[2] *Leo Pharma A/S and another v Sandoz Ltd* [2009] EWHC 996 (Pat), 15 May 2009. `https://www.bailii.org/ew/cases/EWHC/Patents/2009/996.html`.

2.3 Grounds of invalidity

Section 72(1) of the Patents Act 1977 sets out the statutory grounds for patent revocation. This paper focuses on two grounds of invalidity: insufficiency and lack of inventive step (obviousness).

Before discussing these grounds, I define three terms: the priority date, prior art, and the person skilled in the art (UK Intellectual Property Office, nd, secs. 2, 3, 5, and 14).

The *priority date* is the cut-off date used to decide what earlier information can be used against an invention. It is usually the filing date of the patent application.

Prior art is information that was already available to the public before the priority date. It does not have to be an earlier patent or even something described as an invention. It may be a journal article, a product sold to the public, a conference presentation, a public demonstration, a website, or any other public disclosure anywhere in the world.

The *person skilled in the art* is a hypothetical person who is competent in the relevant technical field but is not unusually inventive.

1. **Lack of inventive step (obviousness).** A claim is invalid if the claimed invention would have been obvious to a person skilled in the art, given the relevant prior art and common general knowledge at the priority date.

2. **Insufficiency.** The patent must disclose the invention clearly and completely enough for a person skilled in the art, using common general knowledge, to perform it across the scope of the claim without undue burden.

In this paper, I treat each obviousness attack against a claim separately. A claim may face several attacks and survive some while failing others. One successful attack is sufficient to invalidate the claim.

3 Data collection and description

3.1 Sample scope and judgment collection

I construct a dataset of cases concerning patentability or validity decided by the Patents Court of the High Court of Justice of England and Wales and the Intellectual Property Enterprise Court (IPEC) between 1995 and August 2026.

Scope of the sample. The sample covers three types of proceedings: applications for revocation; infringement proceedings involving a counterclaim challenging patent validity; and appeals to the Patents Court from UKIPO decisions. The third category includes appeals against the refusal of a patent application and appeals concerning the revocation of a granted patent.

I exclude judgments of the Court of Appeal and the UK Supreme Court to avoid counting the same underlying dispute at successive stages, with substantially overlapping patents, parties, and

grounds of attack. A patent may nevertheless appear in separate proceedings when the claims challenged or the prior art supporting an obviousness challenge differ materially.

Sources and inclusion criteria. I identify cases using vLex, a legal research database, and cross-check its listings against BAILII, the British and Irish Legal Information Institute's publicly accessible repository of legal decisions. Some vLex entries contain the full judgment, while others provide only the case title or bibliographic information. When neither source provides the full judgment, I search CaseMine and other available sources.

Where a full judgment is missing, I reconstruct information using a subsequent Court of Appeal judgment. The appellate judgment serves only as a source about the earlier proceedings; the observation and recorded outcome remain those of the Patents Court or IPEC.

A case is retained in the collected dataset only if the available material identifies the parties, the patent or application in dispute, and the outcome. For 1995–2003, at most four otherwise eligible cases are missing in any given year. Coverage of eligible cases is complete for 2004–August 2026.

I then restrict the analysis sample to cases involving at least one insufficiency or obviousness attack. Overall, the combined court sample contains 1,557 claim-level observations.

Once a judgment has been obtained, I use four structured LLM prompts to extract outcome labels and prediction inputs. Judgments are lengthy and often discuss issues beyond the validity challenges studied here, so this step identifies and organizes the relevant information. The full prompts are provided in Appendix D. Table 1 summarizes their inputs and functions.

Table 1: Overview of the extraction prompts

No.	Input	Main purpose	Summary
1	Judgment	Extract case metadata and information needed to construct numerical features.	Records court, parties, dates, patent details, and patent-level outcomes, and retrieves citation information.
2	Judgment, claim texts, and initial extraction records	Identify challenged claims, grounds of attack, and outcomes.	Produces attack-level records, links novelty and obviousness attacks to prior-art units, and records claim dependencies and expanded claim text.
3	Patent publication	Clean the patent text.	Verifies the publication and version, then extracts the English description and claims separately without translation or paraphrasing. Removes bibliographic material, repeated headers and footers, non-English duplicates, tables, drawings, and unusable formula or chemical-structure text.
4	Judgment and prior-art references	Clean the prior-art text.	Verifies the cited document and version and applies the same cleaning rules. Retains full documents or relevant chapters rather than only judge-selected passages. Uses English sources where available and literal translation otherwise.

Below I provide an example of the output of the second prompt. In *Leo Pharma v Sandoz* (Example 2.1), each of claims 1–4 faced one novelty attack based on the patent application WO1991012807A1 and two obviousness attacks based separately on applications WO1991012807A1 and WO1987000834A1. The novelty attack alleged that following Example 4 would inevitably produce a cream containing calcipotriol monohydrate; the obviousness attacks relied on routine formulation and crystallisation in light of the prior art and common general knowledge. The second extraction prompt therefore outputs four novelty and eight obviousness attack-level observations, with none for the other validity grounds.

3.2 Units of observation and outcome coding

I include claims when their validity or patentability is challenged by either insufficiency or obviousness. Claims mentioned only as technical background, through dependency references, or in relation to infringement are not included on that basis alone. Individual claim outcomes are not inferred solely from patent revocation.

Original and amended claim versions are recorded separately when both are assessed; otherwise, only the assessed version is included. A claim survives only if it survives against all attacks and is invalidated otherwise.

3.3 Claim dependencies and expanded claim representations

A dependent claim incorporates the limitations of one or more earlier claims and therefore generally cannot be interpreted from its own text alone. For a given target claim, its dependency structure can be visualized by unfolding the dependency paths into a tree. The target claim forms the root, each branch follows a reference to an earlier claim, and each path terminates at an independent claim. An earlier claim may appear on more than one branch when different dependency paths lead to it.

Definition 3.1 (Maximum dependency depth)**.** A claim's *maximum dependency depth* is the number of dependency links in the longest path from that claim to an independent claim. An independent claim therefore has maximum dependency depth zero. For a dependent claim, the depth equals one plus the greatest maximum dependency depth among its direct parent claims.

Formally, let $\mathcal{P}(c)$ denote the set of direct parent claims referenced by claim c, including all alternatives specified by references such as "any preceding claim." The maximum dependency depth of claim c, denoted by $d(c)$, is

$$d(c) = \begin{cases} 0, & \text{if } \mathcal{P}(c) = \varnothing, \\ 1 + \max\limits_{p \in \mathcal{P}(c)} d(p), & \text{if } \mathcal{P}(c) \neq \varnothing. \end{cases}$$

An *expanded claim* is a self-contained representation that explicitly incorporates the limitations inherited through this dependency structure. The claim-level extraction prompt instructs the LLM to trace every dependency path from the target claim to an independent claim and incorporate the relevant wording along that path. Where the dependency references specify alternative paths, the expanded text preserves those alternatives. Each alternative includes both the limitations inherited along its path and the limitations added by the target claim. For example, the expanded version of claim 5 in Example 2.1 is:

> A pharmaceutical composition containing calcipotriol monohydrate, under any one of the following alternatives: (i) no additional limitation as to dosage form; (ii) the pharmaceutical composition is a cream; or (iii) the pharmaceutical composition is a gel; with an active-component content of 1–100 μg/g of the composition.

Greater dependency depth requires information to be combined across longer chains of claims, potentially increasing the difficulty of interpretation and prediction. I therefore conduct the main specification search and architecture selection using claims with a maximum dependency depth of at most two. After selecting the architectures, I evaluate the same model specifications on the full claim sample without imposing a dependency-depth restriction. This exercise assesses whether architectures selected using claims with simpler dependency structures also perform well when deeper claims are included, without conducting a separate architecture search for the unrestricted sample.

3.4 Training, validation, and evaluation protocol

I select models and architectures using an approximately 80/20 training–validation split of the 1995–2022 court sample. I then evaluate the shortlisted specifications on two holdout sets. The temporal holdout contains court cases decided in 2023–August 2026. It tests whether models trained on past decisions predict outcomes in future disputes.

The institutional holdout contains UKIPO cases. These proceedings are decided by hearing officers rather than judges, providing a test of generalization across institutions and decision-makers. I exclude eight UKIPO cases (eight patent observations) whose Patents Court appeals are already included in the training and validation sample.

3.5 Summary statistics

Table 2 reports claim-level sample sizes and owner-win shares across the development and holdout samples. The depth restrictions are defined in Subsection 3.3; the development and holdout splits are described in Subsection 3.4. As shown in the table, patent owners have a substantially higher win rate at the UKIPO than in court. This pattern potentially reflects structural differences between the two forums.

Table 2: Claim-level validity: sample sizes and owner-win shares

	Training–validation		Court holdout		UKIPO holdout	
Depth	N	Win (%)	N	Win (%)	N	Win (%)
≤ 2	987	41.3	133	54.1	172	72.7
Full	1,374	40.2	183	45.4	293	75.8

4 Models and evaluation metrics

4.1 Embedding, frozen models and Lora-tuning

Before going into my architecture, I introduce several terminologies commonly used in machine learning.

Embeddings. An embedding is a numerical representation of text. A tokenizer first divides the text into tokens drawn from a predefined vocabulary. A token may represent a whole word, part of a word, or punctuation. For example, depending on the tokenizer, "running" might be divided into "run" and "ning".

For a document containing n tokens, a text encoder produces a matrix of contextual token

representations,

$$H = \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix} \in \mathbb{R}^{n \times D},$$

where $h_i \in \mathbb{R}^{1 \times D}$ represents token i in its textual context. The matrix therefore has one row per token and D columns, corresponding to the encoder's hidden dimension.

A pooling operation converts these token representations into a single vector representing the document:

$$e = \text{Pool}(H) \in \mathbb{R}^{1 \times D}.$$

For pooling based on a weighted sum, this operation can be written as

$$e = a^\top H = \sum_{i=1}^{n} a_i h_i, \qquad a \in \mathbb{R}^n.$$

For example, mean pooling sets $a_i = 1/n$, whereas last-token pooling selects the final non-padding token's contextual representation. The resulting document embedding has a fixed dimension regardless of the number of tokens.

To illustrate the intuition, consider two descriptions inspired by Example 2.1: "A cream containing calcipotriol monohydrate" and "Calcipotriol monohydrate formulated as a cream." Although their wording differs, they express similar content. A useful semantic embedding should therefore place them close together in the representation space, as illustrated in Figure 1.

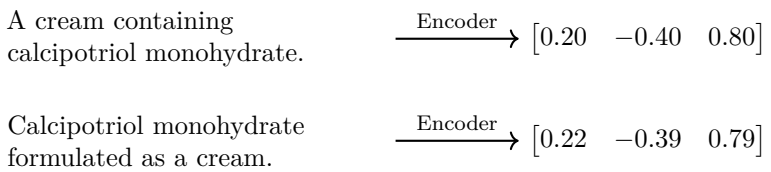


Figure 1: Illustration of semantic text embeddings. The encoder includes document pooling. The three-dimensional vectors are illustrative and are not actual encoder outputs.

Frozen embeddings One approach uses a pretrained text encoder with its parameters held fixed. In this paper, I obtain frozen document embeddings from `Qwen3-Embedding-8B` (Zhang et al., 2025), which uses last-token pooling. The encoder is not updated using the outcome labels. Only the downstream layers, including any projection layers and the classification head, are trained for the prediction task.

LoRA tuning. Another approach is to retrain the encoding. Fine-tuning adapts those representations to a particular task using additional training data. Full fine-tuning updates all encoder parameters alongside the downstream prediction layers. This can be computationally expensive and raises concerns about overfitting when labeled observations are scarce relative to the number of

parameters like in our case.

I therefore use Low-Rank Adaptation (LoRA) (Hu et al., 2022). LoRA keeps the pretrained weights fixed and learns low-rank updates to selected weight matrices. For a pretrained matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the adapted matrix is

$$W = W_0 + \frac{\alpha}{r} BA,$$

where

$$A \in \mathbb{R}^{r \times d_{\text{in}}}, \qquad B \in \mathbb{R}^{d_{\text{out}} \times r}.$$

Within each adapted weight matrix, LoRA keeps W_0 fixed and trains only A and B. It therefore updates $r(d_{\text{in}} + d_{\text{out}})$ parameters, compared with the $d_{\text{out}} \times d_{\text{in}}$ parameters required to update the full matrix.

The rank r controls the number of trainable parameters and the maximum rank of the adjustment. Increasing r allows more flexible adaptation. The parameter α sets the scaling factor α/r: at a fixed rank, it controls the contribution of a given low-rank update without changing the number of trainable parameters.

4.2 Claim-level validity across multiple attacks

The frozen and LoRA-tuned encoders described above provide numerical representations of the relevant legal texts. To predict claim validity, these representations must be combined across the attacks brought against each claim. I formulate this task as a multiple-instance learning problem: each claim forms a bag, and each attack against that claim corresponds to one instance. Every instance contains the expanded claim text c, the full claim set s, and the textual material a_i associated with attack i. A claim facing $K \geq 1$ attacks is therefore represented by

$$X = \{(c, s, a_i)\}_{i=1}^{K}.$$

For insufficiency, a_i is the patent description. For obviousness, it contains the prior-art document or documents assessed jointly as one attack. All instances in a bag share the claim-side texts c and s, while their attack-specific material may differ.

Let $y_i = 1$ if attack i succeeds and $y_i = 0$ otherwise. The claim is invalid if at least one attack succeeds, giving the claim-level label

$$y = \begin{cases} 0, & \sum_{i=1}^{K} y_i = 0, \\ 1, & \sum_{i=1}^{K} y_i > 0. \end{cases} \tag{1}$$

This rule defines the target label. The models differ in how they aggregate information across attacks to predict it.

I compare three supervised approaches:

- **Individual-attack prediction (IND).** I first select the best-performing attack-level prediction model using validation data. Its predicted success probabilities are then aggregated across the attacks against each claim to obtain the bag-level probability. The aggregation methods are either mean or max pooling (see Appendix B.2).
- **Attention-based MIL (ILSE).** Following Ilse et al. (2018), instance representations are aggregated using learned attention weights. I consider both attention and gated-attention pooling when selecting the ILSE baseline (see B.2).
- **Claim-query cross-attention (CUS).** In my proposed architecture, the representations of the expanded claim and the full claim set are combined into a single query. This query attends to attack representations augmented with ground-type embeddings.

IND and ILSE serve as baselines for our model.

Attention blocks. Let U, V be matrices with the same number of columns. I combine multi-head attention, residual connections, layer normalization, and a row-wise feedforward network, as in the Transformer (Vaswani et al., 2017). The resulting multi-head attention block is denoted by MAB:

$$H = \mathrm{LayerNorm}\big(U + \mathrm{Multihead}(U, V, V)\big), \tag{2}$$

$$\mathrm{MAB}(U, V) = \mathrm{LayerNorm}\big(H + \mathrm{rFF}(H)\big), \tag{3}$$

where rFF is a feedforward transformation applied independently to each row, with shared parameters. The first argument supplies the queries, while the second supplies the keys and values.

Self-attention is obtained by using the same representations for the queries, keys, and values:

$$\mathrm{SAB}(U) = \mathrm{MAB}(U, U). \tag{4}$$

This self-attention block allows the rows of U to interact. LayerNorm, multihead attention, and rFF are all mathematical operators. Appendix B.1 defines LayerNorm, multihead attention, and rFF and explains their roles in the block.

Text representations. I encode the expanded claim, full claim set, and attack-specific texts separately. The claim-side encoder is adapted with LoRA, while the encoders for the description and prior-art texts remain frozen. The reason is that the description and the prior arts are usually much longer, using full Lora-tuning can lead to overfit. The resulting embeddings are projected to a common dimension d.

Let $e_c, e_s \in \mathbb{R}^{1\times d}$ denote the projected embeddings of the expanded claim and full claim set, respectively. Let $A_i \in \mathbb{R}^{1\times d}$ denote the projected embedding of the textual material a_i associated with attack i.

Claim-side interaction and fusion. The two claim-side representations are stacked as

$$U = \begin{bmatrix} e_c \\ e_s \end{bmatrix} \in \mathbb{R}^{2\times d}.$$

I apply self-attention to the two claim-side representations:

$$\bar{U} = \mathrm{SAB}(U) = \begin{bmatrix} \bar{e}_c \\ \bar{e}_s \end{bmatrix} \in \mathbb{R}^{2\times d},$$

where $\bar{e}_c$ and $\bar{e}_s$ are the updated representations of the expanded claim and the full claim set, respectively. Self-attention allows each representation to incorporate information from the other while preserving the matrix's two-row, d-column shape.

I then flatten $\bar{U}$ by placing its two rows side by side to form a single row vector:

$$u = [\bar{e}_c, \bar{e}_s] \in \mathbb{R}^{1\times 2d}.$$

The fusion layer combines these entries through multiplication by a learned matrix W_{fuse} and addition of a bias b_{fuse}:

$$v = uW_{\text{fuse}} + b_{\text{fuse}}, \qquad W_{\text{fuse}} \in \mathbb{R}^{2d\times d}, \qquad b_{\text{fuse}} \in \mathbb{R}^{1\times d}.$$

Finally, I apply ReLU and dropout:

$$q = f_{\text{fuse}}(u) = \mathrm{Dropout}(\mathrm{ReLU}(uW_{\text{fuse}} + b_{\text{fuse}})) \in \mathbb{R}^{1\times d}.$$

The resulting vector q is the single claim query used in the subsequent cross-attention over attacks. Definitions of Dropout and RELU are in the Appendix B.1.

Attack representations and legal grounds. The attack representations are stacked as

$$V = [A_1; \ldots; A_K] \in \mathbb{R}^{K\times d},$$

with one row for each attack.

To provide explicit information about the legal ground of each attack, I add learned ground-type embeddings. Let $g_i \in \{\mathrm{INS}, \mathrm{OBV}\}$ denote the ground of attack i, and let $t_{\mathrm{INS}}, t_{\mathrm{OBV}} \in \mathbb{R}^{1\times d}$ be learned vectors. The augmented attack representations are

$$\widetilde{A}_i = A_i + t_{g_i}, \qquad i = 1, \ldots, K. \tag{5}$$

Let

$$\widetilde{V} = [\widetilde{A}_1; \ldots; \widetilde{A}_K] \in \mathbb{R}^{K\times d}.$$

These embeddings distinguish insufficiency from obviousness within the common representation space.

Claim-query cross-attention and prediction. The fused claim query interacts with the augmented attack representations through

$$Z = \mathrm{MAB}(q, \widetilde{V}) \in \mathbb{R}^{1\times d}, \tag{6}$$

where q supplies a single query and $\widetilde{V}$ supplies the keys and values.

The motivation is that the relevance of an attack depends on the claim being challenged. Using the fused claim representation as the query allows attention weights over attacks to depend explicitly on both the expanded claim and the full claim set. Separate encoding preserves the distinction between claim-side information shared across attacks and the textual material specific to each attack.

The output Z is concatenated with numerical features, and passed to a classification head to predict claim invalidity. Appendix A provides the full list of numerical features. The complete CUS architecture is illustrated in Figure C.1 in Appendix C.1.

4.3 Evaluation metrics

Because outcome classes are often imbalanced, accuracy alone can be misleading. I select models using validation-set Matthews correlation coefficient (MCC), with macro-F1 and then accuracy as secondary criteria when MCC values are very similar. Intuitively, MCC measures the correlation between the model's predictions and the ground truth. Formally, let TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. Then:

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}.$$

MCC equals 1 for perfect prediction, 0 when the predictions have no overall association with the observed outcomes, and -1 for perfect disagreement. Because MCC takes into account all the four elements of the confusion matrix, MCC is my principal evaluation metric. Normally, an observed MCC of over 0.2 suggests that the model captures some predictive signal.

5 Results

Table 3 reports MCC, macro-F1, and accuracy on the court and UKIPO holdouts, with the two samples shown side by side. Definitions of the model labels are collected in Table A.2; Subsection 4.2 describes the architectures. The zero-shot ChatGPT benchmark uses the prompt reproduced in Appendix D.5.

Table 3: Claim-level validity on the court and UKIPO holdouts

	Court holdout			UKIPO holdout		
Model	MCC	Macro-F_1	Accuracy	MCC	Macro-F_1	Accuracy
Panel A: Depth ≤ 2		$n = 133$			$n = 172$	
IND (mean)	0.124	0.554	0.571	-0.227	0.369	0.407
ILSE	0.320	0.634	0.639	0.178	0.587	0.651
CUS	**0.457**	**0.722**	**0.722**	**0.291**	**0.643**	**0.733**
ChatGPT	0.149	0.499	0.534	-0.030	0.395	0.395
Panel B: Full depth		$n = 183$			$n = 293$	
IND (mean)	0.154	0.577	0.579	-0.195	0.316	0.317
ILSE	0.194	0.564	0.607	-0.009	0.456	0.488
CUS	**0.334**	**0.616**	**0.661**	**0.084**	**0.525**	**0.584**
ChatGPT	0.195	0.561	0.607	-0.139	0.333	0.334

CUS has the highest MCC, macro-F1, and accuracy among the reported models in both holdouts and at both depth levels. On the court holdout, its MCC is 0.457 for depth ≤ 2 and 0.334 for full depth; on UKIPO, the corresponding values are 0.291 and 0.084. Thus, its relative advantage persists across institutions, but its full-depth UKIPO performance remains weak.

Bootstrap assessment of MCC differences. To assess uncertainty in the test-set performance comparisons, I conduct a paired nonparametric bootstrap over claims (Efron and Tibshirani, 1993). For each combination of dependency depth and test set containing N claims, I generate $B = 10{,}000$ bootstrap samples by drawing N claims with replacement from the original test set. A claim may therefore appear multiple times in a bootstrap sample, while others may not appear. The same sampled claims and their multiplicities are used for all three models, and predictions remain fixed throughout. For the combined test set, sampling is stratified by source, preserving the original numbers of Court and UKIPO claims.

In each bootstrap replicate b, I calculate the MCC difference between CUS and each comparator:

$$\Delta_m^{(b)} = \mathrm{MCC}_{\mathrm{CUS}}^{(b)} - \mathrm{MCC}_m^{(b)}, \qquad m \in \{\mathrm{ILSE}, \mathrm{ChatGPT}\}.$$

For a confidence level of $1 - \alpha$, the percentile confidence interval is given by the empirical $\alpha/2$ and $1 - \alpha/2$ quantiles of the bootstrap differences. I report intervals at the 90%, 95%, and 99% confidence levels. An interval excluding zero indicates statistical significance at the corresponding nominal two-sided level under this bootstrap procedure.

Table 4 reports the MCC differences on the original test sets and their bootstrap confidence intervals. The estimated differences favor CUS in all twelve comparisons. The 90%, 95%, and 99% intervals exclude zero in eleven, nine, and six comparisons, respectively.

Table 4: Paired claim-bootstrap confidence intervals for MCC differences

Test set	Comparator	$\widehat{\Delta}$	90% CI	95% CI	99% CI
Panel A: Depth ≤ 2					
Court ($N = 133$)	ILSE	0.137	[0.018, 0.255]	[−0.005, 0.279]	[−0.049, 0.325]
	ChatGPT	0.308	[0.131, 0.489]	[0.095, 0.524]	[0.031, 0.596]
UKIPO ($N = 172$)	ILSE	0.113	[0.023, 0.200]	[0.005, 0.218]	[−0.028, 0.255]
	ChatGPT	0.322	[0.122, 0.515]	[0.087, 0.555]	[0.018, 0.625]
Full ($N = 305$)	ILSE	0.123	[0.056, 0.189]	[0.044, 0.201]	[0.020, 0.225]
	ChatGPT	0.344	[0.223, 0.469]	[0.198, 0.493]	[0.151, 0.542]
Panel B: Full Depth					
Court ($N = 183$)	ILSE	0.140	[0.042, 0.240]	[0.023, 0.258]	[−0.013, 0.295]
	ChatGPT	0.139	[−0.022, 0.301]	[−0.054, 0.337]	[−0.114, 0.395]
UKIPO ($N = 293$)	ILSE	0.093	[0.013, 0.173]	[−0.004, 0.187]	[−0.031, 0.216]
	ChatGPT	0.223	[0.077, 0.366]	[0.048, 0.391]	[−0.004, 0.458]
Full ($N = 476$)	ILSE	0.133	[0.075, 0.192]	[0.064, 0.204]	[0.042, 0.229]
	ChatGPT	0.256	[0.159, 0.356]	[0.142, 0.373]	[0.107, 0.407]

Notes: $\widehat{\Delta}$ is the original test-set MCC of CUS minus that of the comparator; positive values favor CUS. N denotes the number of claims. Full combines Court and UKIPO. Intervals use 10,000 paired claim-bootstrap replicates.

For claims with a dependency depth of at most two, CUS achieves higher MCC than both ChatGPT and ILSE across all three test sets, with the 90% bootstrap confidence intervals for the differences lying entirely above zero. The corresponding 95% intervals also exclude zero, except for the comparison with ILSE on the Court test set. For the full-depth sample, the evidence is somewhat weaker, although the 90% intervals remain entirely above zero in all comparisons except CUS versus ChatGPT on the Court test set.

6 Attention weights and explainability

This section examines the explanatory role of attention weights in my model. The literature distinguishes between *plausibility*, which concerns whether an explanation agrees with human judgments, and *faithfulness*, which concerns whether it accurately reflects the process that produces the model's prediction (Bibal et al., 2022). Jain and Wallace (2019) show that, in the models they study, substantially different attention distributions can produce nearly identical predictions. These findings challenge the interpretation of attention weights as measures of input importance. Wiegreffe and Pinter (2019), however, argue that such results do not generally rule out the explanatory value of attention.

Related evaluations include the attention-erasure experiments of Serrano and Smith (2019) and the comprehensiveness and sufficiency metrics in ERASER (DeYoung et al., 2020). For an overview of the debate, see Bibal et al. (2022). Erasure-based evaluations require care in my setting. First, removing a highly weighted input may have little effect when other inputs contain overlapping predictive signals. Moreover, deleting an entire attack can change the claim-level target under the

task's aggregation rule. Second, more importantly, these studies do not provide a statistical test for comparing faithfulness across models.

I therefore propose a regression-based sensitivity diagnostic. Motivated by the logic of randomized interventions (Duflo et al., 2007), I replace the attention distributions while keeping the input texts, numerical features, and all trained parameters fixed. I then recompute the downstream predictions without retraining. This preserves the complete set of attacks and measures the model's response to alternative allocations of attention.

I apply this diagnostic to CUS, which has four cross-attention heads. Let i index claims, $h \in \{1, 2, 3, 4\}$ index heads, and n_i denote the number of attacks on claim i. The analysis includes every test claim with $n_i \geq 2$, regardless of whether its original prediction is correct. Let $\boldsymbol{\alpha}_{ih}$ denote the original attention vector over its attacks. For each claim, I generate $R = 400$ replications, drawing fresh attention vectors independently across heads and replications:

$$\widetilde{\boldsymbol{\alpha}}_{ihr} \sim \text{Dirichlet}(\underbrace{1, \ldots, 1}_{n_i}), \qquad r = 1, \ldots, R. \tag{7}$$

This distribution is uniform over the simplex, so the replacement weights are nonnegative and sum to one within each head. All four heads are replaced in each replication.

Let p_i be the original predicted probability of claim invalidity and $\widetilde{p}_{ir}$ the probability after the intervention. The dependent variable is the absolute change in log-odds:

$$Y_{ir} = \left| \log\left(\frac{\widetilde{p}_{ir}}{1 - \widetilde{p}_{ir}}\right) - \log\left(\frac{p_i}{1 - p_i}\right) \right|. \tag{8}$$

The first specification measures the overall attention change by the root mean square of the four head-level Euclidean distances:

$$D_{ir} = \left[\frac{1}{4} \sum_{h=1}^{4} \sum_{j=1}^{n_i} \left(\widetilde{\alpha}_{ihjr} - \alpha_{ihj}\right)^2 \right]^{1/2}, \tag{9}$$

where j indexes attacks and $\widetilde{\alpha}_{ihjr}$ is the jth element of $\widetilde{\boldsymbol{\alpha}}_{ihr}$. I estimate

$$Y_{ir} = \mu_i + \beta D_{ir} + \varepsilon_{ir}, \tag{10}$$

where μ_i is a claim fixed effect. A positive β indicates that, within the same claim, larger attention reallocations are associated on average with larger absolute changes in predicted log-odds. An increase of 0.1 in D_{ir} corresponds to an estimated increase of 0.1β in the absolute log-odds change.

The second specification distinguishes the four heads. For each head, define

$$d_{ir}^{(h)} = \left\| \widetilde{\boldsymbol{\alpha}}_{ihr} - \boldsymbol{\alpha}_{ih} \right\|_2 = \left[\sum_{j=1}^{n_i} \left(\widetilde{\alpha}_{ihjr} - \alpha_{ihj}\right)^2 \right]^{1/2}. \tag{11}$$

I then estimate all four coefficients jointly:

$$Y_{ir} = \mu_i + \sum_{h=1}^{4} \beta_h d_{ir}^{(h)} + \varepsilon_{ir}. \tag{12}$$

Each β_h measures the association between the distance for head h and the prediction change, conditional on the distances for the other heads. Differences between heads can be assessed by testing $H_0 : \beta_h = \beta_k$.

Both specifications are estimated by ordinary least squares with claim fixed effects and standard errors clustered by case (Cameron and Miller, 2015). Clustering allows dependence across replications and across claims belonging to the same case. I estimate the specifications separately for the pooled sample (court and UKIPO), the court sample, and the UKIPO sample. Table 5 reports the aggregate-distance and joint four-head regressions for the pooled, court, and UKIPO samples. Panel A presents depth ≤ 2, and Panel B presents full depth.

Table 5: Attention reallocation and prediction sensitivity

	Pooled		Court		UKIPO	
	Aggregate (1)	By head (2)	Aggregate (3)	By head (4)	Aggregate (5)	By head (6)
Panel A: Depth ≤ 2						
Aggregate distance D_{ir}	0.04074***		0.04320***		0.03632**	
	(0.00991)		(0.01276)		(0.01526)	
Head 1 distance $d_{ir}^{(1)}$		0.00905***		0.01013***		0.00714*
		(0.00248)		(0.00319)		(0.00362)
Head 2 distance $d_{ir}^{(2)}$		0.00541***		0.00604***		0.00430**
		(0.00142)		(0.00193)		(0.00182)
Head 3 distance $d_{ir}^{(3)}$		0.00886***		0.00942***		0.00788*
		(0.00217)		(0.00256)		(0.00388)
Head 4 distance $d_{ir}^{(4)}$		0.01402***		0.01371***		0.01455**
		(0.00324)		(0.00421)		(0.00523)
Observations	68,400	68,400	43,200	43,200	25,200	25,200
Claims	171	171	108	108	63	63
Cases (clusters)	52	52	30	30	22	22
Claim fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Within R^2	0.0493	0.0501	0.0571	0.0553	0.0373	0.0427
Panel B: Full depth						
Aggregate distance D_{ir}	0.02417***		0.02561***		0.02223***	
	(0.00556)		(0.00769)		(0.00776)	
Head 1 distance $d_{ir}^{(1)}$		0.00434***		0.00457***		0.00404***
		(0.00101)		(0.00141)		(0.00141)
Head 2 distance $d_{ir}^{(2)}$		0.00785***		0.00875***		0.00665**
		(0.00174)		(0.00224)		(0.00256)
Head 3 distance $d_{ir}^{(3)}$		0.00533***		0.00534***		0.00531***
		(0.00128)		(0.00178)		(0.00185)
Head 4 distance $d_{ir}^{(4)}$		0.00458***		0.00492**		0.00412**
		(0.00127)		(0.00189)		(0.00151)
Observations	102,800	102,800	58,800	58,800	44,000	44,000
Claims	257	257	147	147	110	110
Cases (clusters)	53	53	31	31	22	22
Claim fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Within R^2	0.0270	0.0262	0.0222	0.0221	0.0447	0.0420

Notes: The dependent variable is the absolute change in predicted log-odds. Columns (1), (3), and (5) use aggregate attention distance; columns (2), (4), and (6) jointly include the four head-specific distances. Case-clustered standard errors are in parentheses. Stars use unadjusted two-sided t-tests with $G-1$ degrees of freedom, where G is the number of cases: $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$.

Table 5 shows positive aggregate coefficients in every sample. For full depth, all three coefficients are statistically significant at the 1% level. For depth ≤ 2, the pooled and court coefficients are significant at the 1% level, while the UKIPO coefficient is significant at the 5% level ($p = 0.0269$). Table 5 shows that all four head coefficients are positive and significant at the 1% level in the pooled

sample for both depths. Significance varies across heads in the venue-specific regressions.

These results provide evidence that the allocation of attention systematically affects the model's predictions under the specified intervention distribution.

7 Conclusion

This paper predicts claim-level patent validity conditional on the insufficiency and obviousness attacks brought against each claim. The dataset links judicial decisions to individual claims, grounds of challenge, and prior-art units. Expanded claim representations make inherited limitations explicit.

My main architecture (CUS) achieves the highest MCC, macro-F1, and accuracy among the evaluated models, including ChatGPT and the MIL baselines, on both the court and UKIPO holdouts at both depth levels. The low MCC on the full-depth UKIPO sample nevertheless indicates limited generalization across institutional settings, with weak performance across all evaluated models in this setting.

I also propose a new test of attention faithfulness that provides statistical inference on the effects of attention reallocation. Holding inputs and trained model parameters fixed, I show that changes in attention allocation affect the predictions of CUS.

The first extension to this paper is to test whether pre-judgment predictions contain trading signals not already reflected in stock prices. The present U.K. sample contains too few relevant litigation events to reliably assess whether a strategy based on these predictions would be profitable after transaction costs.

The second extension is to compare predicted vulnerability to insufficiency between challenged and otherwise similar unchallenged claims, examining whether challengers target claims that the model also identifies as weaker. Interpreting these comparisons requires accounting for selection into challenges, since outcomes for unchallenged claims are unobserved.

A Numerical Features and Model Labels

A.1 Numerical features

Table A.1: Definition and use of numerical features

Feature	Definition
`claimant_entity`	Legal status of the claimant. It equals 1 if the claimant is an individual, a private company, or a non-profit organization such as a university; 2 if the claimant is a publicly listed company; and 3 if the claimant is a public authority, specifically the Comptroller.
`defendant_entity`	Legal status of the defendant. It equals 1 if the defendant is an individual, a private company, or a non-profit organization such as a university; 2 if the defendant is a publicly listed company; and 3 if the defendant is a public authority, specifically the Comptroller.
`claimant_headquarter`	Location of the claimant's headquarters. It equals 0 for the United Kingdom; 1 for the European Union or Switzerland, excluding the United Kingdom; and 2 for all other countries.
`defendant_headquarter`	Location of the defendant's headquarters. It equals 0 for the United Kingdom; 1 for the European Union or Switzerland, excluding the United Kingdom; and 2 for all other countries.
`patent_age`	Age of the patent, measured in days from the patent's priority date to the date of the first judgment in the case. Formally, it is calculated as the first judgment date minus the priority date.
`forward_citation`	Number of times the patent had been cited by other patents as of the date of the first judgment. Citations occurring after that date are not counted.
`backward_citation`	Number of earlier patents cited by the patent.
`claimant_is_owner`	Indicator for the relationship between the claimant and the patent owner. It equals 0 if the claimant is the patent owner and 1 otherwise.
`amended`	Indicator for whether the claim is an amended version. It equals 0 if the claim was not amended and 1 if it was amended.
`dependency_depth`	Maximum number of dependency links in any path connecting the claim to an independent claim. An independent claim has depth zero. The depth of a dependent claim is one plus the greatest depth among its parent claims, as defined in Definition 3.1.

A.2 Model labels and abbreviations

Table A.2 defines the models retained in the claim-level comparison. In the ILSE and CUS models, the claim-side encoder uses LoRA and the attack-text encoders are frozen; downstream layers remain trainable.

Table A.2: Claim-level model labels and constructions

Label	**Model construction**
IND (mean)	Mean of the individual-attack probabilities from the insufficiency and obviousness classifiers; see Appendix B.2.
ILSE	Each instance combines the expanded claim, complete claim set, and one attack's text. Attention-based MIL pools these representations.
CUS	Self-attention combines the expanded-claim and complete-claim-set representations before fusion into one query. Ground-type embeddings augment the attack keys and values, and claim-query cross-attention aggregates the attacks. The reported configuration is C182; see Figure C.1.
ChatGPT	GPT-5.6 Sol, evaluated zero-shot using the claim-level prompt in Appendix D.5.

B Claim-Level Model Details

B.1 Mathematical operations

This subsection defines the operators used in the in Subsection 4.2.

Layer normalization. Layer normalization (Ba et al., 2016, Section 3) centres and rescales each representation using its own coordinates. For a row with entries $z = (z_1, \ldots, z_p)$, define

$$\mu(z) = \frac{1}{p}\sum_{j=1}^{p} z_j, \qquad s^2(z) = \frac{1}{p}\sum_{j=1}^{p}\big(z_j - \mu(z)\big)^2. \tag{13}$$

The normalized output is

$$\big[\mathrm{LayerNorm}(z)\big]_j = g_j \frac{z_j - \mu(z)}{\sqrt{s^2(z) + \epsilon}} + b_j, \qquad j = 1, \ldots, p, \tag{14}$$

where $\epsilon > 0$ is a small constant for numerical stability, and g_j and b_j are learned scale and shift parameters. For a matrix, the operation is applied separately to every row, with the same learned parameters across rows within each LayerNorm layer. Thus, the mean and variance are computed across features of one representation, not across attacks or observations in a training batch. Intuitively, this controls the level and scale of the representations passed between layers, while allowing the model to learn suitable scales and offsets. The final output need not have zero mean or unit variance because of the learned affine transformation.

Multihead attention. Following Vaswani et al. (2017, Sections 3.2.1–3.2.2), let $U \in \mathbb{R}^{n_q \times d}$ and $V \in \mathbb{R}^{n_v \times d}$.

An attention *head* is one attention computation with its own learned query, key, and value projections. Multihead attention performs h such computations in parallel on the same inputs. Each head can therefore learn a different matching pattern between the query-side and source-side representations.

For each head $r = 1, \ldots, h$, learned projections produce queries, keys, and values:

$$Q_r = UW_r^Q, \qquad K_r = VW_r^K, \qquad R_r = VW_r^V, \tag{15}$$

where $W_r^Q, W_r^K \in \mathbb{R}^{p \times d_k}$ and $W_r^V \in \mathbb{R}^{p \times d_v}$. Bias terms are omitted for notational simplicity. Each projection uses all p input coordinates to construct the representation used by that head.

Under the standard equal-width construction, $d_k = d_v = d/h$, assuming that d is divisible by h.

For example, if $d = 128$, each head uses 32-dimensional queries, keys, and values. Its projection matrices have dimension 128×32, giving

$$Q_r \in \mathbb{R}^{n_q \times 32}, \qquad K_r, R_r \in \mathbb{R}^{n_v \times 32}.$$

Queries specify what information to seek; keys determine how strongly each source matches a query; values supply the information to combine. Each head computes

$$S_r = \frac{Q_r K_r^\top}{\sqrt{d_k}}, \qquad \alpha_{r,ij} = \frac{\exp(S_{r,ij})}{\sum_{\ell=1}^{n_v} \exp(S_{r,i\ell})}, \qquad \text{head}_r = \alpha_r R_r. \tag{16}$$

The softmax is applied across source rows for each query, so its weights are nonnegative and sum to one. Each output row is a weighted combination of value vectors. Scaling by $\sqrt{d_k}$ moderates the magnitude of the dot products. The heads are concatenated along the feature dimension and projected back to p coordinates:

$$\text{Multihead}(U, V, V) = \text{Concat}\big(\text{head}_1, \ldots, \text{head}_h\big) W^O, \qquad W^O \in \mathbb{R}^{hd_v \times d}. \tag{17}$$

with W^O another learned matrix.

Different heads can learn different matching patterns through their separate projection matrices.

Row-wise feedforward transformation (rFF). The abbreviation rFF denotes a feedforward transformation applied separately to each row of a matrix, using the same learned parameters for every row (Lee et al., 2019, Section 3.1). For $H \in \mathbb{R}^{n \times d}$, with rows $h_i \in \mathbb{R}^{1 \times d}$, write

$$\text{rFF}(H) = \begin{bmatrix} f(h_1) \\ \vdots \\ f(h_n) \end{bmatrix}. \tag{18}$$

In my architecture, the shared function f consists of a single learned affine transformation followed by ReLU:

$$f(x) = \text{ReLU}(xW + b), \tag{19}$$

where $W \in \mathbb{R}^{d \times d}$ and $b \in \mathbb{R}^{1 \times d}$. The matrix multiplication combines coordinates within each row, and ReLU is applied element-wise. The same weight matrix W and bias b are used for every row.

Each transformed row retains d coordinates, so $\text{rFF}(H) \in \mathbb{R}^{n \times d}$. This preserves the dimensions required for the residual addition $H + \text{rFF}(H)$ in the attention block.

Rectified linear unit (ReLU). ReLU is an activation function applied separately to every coordinate (Glorot et al., 2011). Let z be any vector:

$$\big[\text{ReLU}(z)\big]_j = \max\{0, z_j\}. \tag{20}$$

It replaces negative entries with zero and leaves nonnegative entries unchanged. For example,

$$\text{ReLU}\Big(\begin{bmatrix} -1.2 & 0 & 0.8 \end{bmatrix}\Big) = \begin{bmatrix} 0 & 0 & 0.8 \end{bmatrix}.$$

ReLU preserves the input dimensions and has no trainable parameters. It introduces nonlinearity to our model.

Dropout. Dropout is a regularization method that randomly masks activation coordinates during training, reducing reliance on particular combinations of features (Srivastava et al., 2014). Let $\rho \in [0, 1)$ be the dropout probability. Under the inverted-dropout convention, its action on a vector z is

$$\left[\text{Dropout}_\rho(z)\right]_j = \frac{m_j}{1-\rho} z_j, \qquad m_j \overset{\text{ind}}{\sim} \text{Bernoulli}(1-\rho). \tag{21}$$

A fresh random mask is drawn on each training pass. An entry with $m_j = 0$ is set to zero; a retained entry is scaled by $1/(1-\rho)$. This scaling preserves each activation's conditional expectation:

$$\mathbb{E}\Big[\left[\text{Dropout}_\rho(z)\right]_j \mid z\Big] = z_j.$$

During validation and test prediction, dropout is disabled, so the operation returns z unchanged. The random masking therefore regularizes training without introducing dropout randomness into the reported test predictions. Dropout preserves the vector's dimensions and adds no trainable parameters. In the reported CUS architecture, the fusion layer and final hidden classification layer use $\rho = 0.1$: each coordinate is masked with probability 10% during training, and retained entries are divided by 0.9.

Connection to claim-query fusion. After the two claim-side rows are flattened into $u \in \mathbb{R}^{1\times 2d}$, the separate fusion layer applies these operations in sequence:

$$q = \text{Dropout}_\rho(\text{ReLU}(uW_{\text{fuse}} + b_{\text{fuse}})) \in \mathbb{R}^{1\times d}, \tag{22}$$

where $W_{\text{fuse}} \in \mathbb{R}^{2d\times d}$ and $b_{\text{fuse}} \in \mathbb{R}^{1\times d}$. Here, the matrix multiplication combines information from both claim-side rows into one query; ReLU supplies the nonlinearity, and dropout regularizes the resulting features during training.

B.2 Multiple-instance pooling

Mean and max probability pooling (IND). Let $p_i \in [0, 1]$ be the predicted probability that attack i succeeds, obtained from the corresponding attack-level model. For a claim facing $K \geq 1$ attacks, mean and max pooling produce the scores

$$\widehat{p}_{\text{mean}} = \frac{1}{K}\sum_{i=1}^{K} p_i, \qquad \widehat{p}_{\text{max}} = \max_{1\leq i\leq K} p_i. \tag{23}$$

The claim is predicted to be invalid when the chosen score $\widehat{p}$ is at least 0.5. Mean pooling averages the predicted attack probabilities, whereas max pooling uses the largest probability. These operators pool probabilities rather than text embeddings.

Attention-based MIL pooling. Following Ilse et al. (2018), let $h_i \in \mathbb{R}^m$ be the representation of instance i. In this application, each instance combines the shared claim-side representations with its attack-specific representation. The pooled bag representation is

$$b = \sum_{i=1}^{K} \alpha_i h_i, \qquad \alpha_i = \frac{\exp(s_i)}{\sum_{j=1}^{K} \exp(s_j)}. \tag{24}$$

The attention weights are nonnegative and sum to one. For the attention operator without a gate, the score is

$$s_i = w^\top \tanh(W h_i), \tag{25}$$

where $W \in \mathbb{R}^{\ell \times m}$ and $w \in \mathbb{R}^\ell$ are learned parameters. The pooled representation b is concatenated with numerical features then passed to a classification head.

Gated-attention MIL pooling. The gated variant of Ilse et al. (2018) replaces the score in Equation (25) with

$$s_i = w^\top \left[\tanh(W h_i) \odot \sigma(R h_i)\right], \tag{26}$$

where $R \in \mathbb{R}^{\ell \times m}$ is an additional learned matrix, σ is the element-wise sigmoid function, and $\odot$ denotes element-wise multiplication. The sigmoid gate modulates each component of the transformed instance representation before its attention score is computed. The normalization and pooling in Equation (24) remain unchanged.

B.3 Macro-F1 and accuracy

Let TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives. The positive outcome is invalidity on at least one retained attack.

Macro-F1. For class c, the F1 score is the harmonic mean of precision and recall and can be written as

$$\mathrm{F1}_c = \frac{2TP_c}{2TP_c + FP_c + FN_c},$$

where TP_c, FP_c, and FN_c are computed by treating class c as the positive class. I report macro-F1,

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{2}\left(\mathrm{F1}_0 + \mathrm{F1}_1\right),$$

which gives equal weight to predictive performance on the two outcome classes.

Accuracy. Accuracy is the fraction of observations assigned to the correct class:

$$\mathrm{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Accuracy provides an intuitive measure of the overall proportion of correct predictions. However, it can be misleading when one class is substantially more frequent because a classifier can obtain relatively high accuracy by predominantly predicting the majority class.

C CUS Architecture

C.1 CUS model architecture

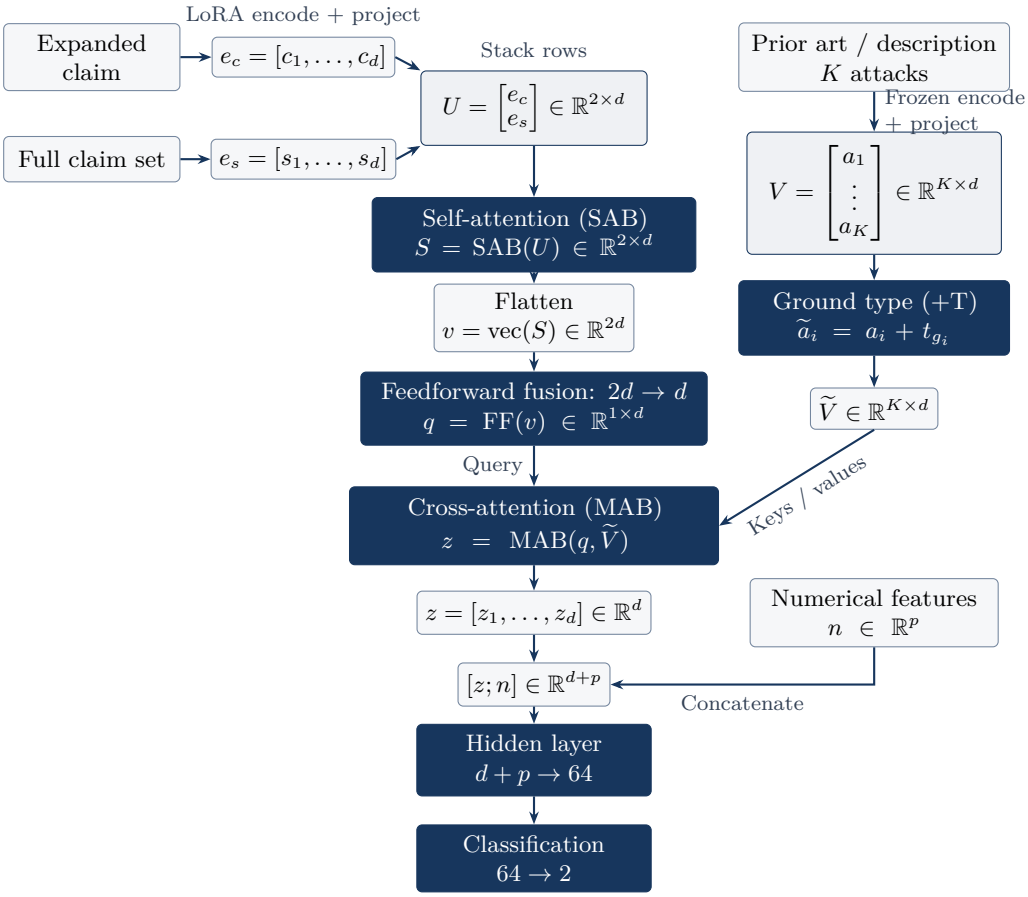


Figure C.1: Full architecture of CUS (C182). The displayed text embeddings are projected to $d = 256$ dimensions; $p = 10$ denotes the number of numerical features. The two claim texts share a LoRA-adapted encoder, while the attack-text encoders are frozen. Each attack contributes one row: a patent description for insufficiency or a jointly packed prior-art unit for obviousness. After SAB, the two rows are flattened to $2d$ coordinates and passed through a separate fusion FF: Linear$(2d, d)$, ReLU, and dropout with rate 0.1. SAB and MAB include their internal feedforward, residual, and LayerNorm operations. The final hidden layer uses ReLU and dropout with rate 0.1, followed by a linear map to two class logits.

D LLM Prompts

The following four subsections reproduce the extraction prompts for patent-level records, claim-level validity attacks, patent text, and prior-art text. The final subsection reproduces the claim-level zero-shot prediction prompt. Source wording, examples, and output instructions are retained; formatting is adapted for the appendix.

D.1 First prompt: patent-level extraction

IP JUDGMENT EXTRACTION PROMPT — PATENT ONLY — RIGHT-LEVEL VALIDITY DATASET

Right-level UK patent-validity dataset — revised with a mandatory Google Patents visual citation example

You are extracting structured data from UK patent judgments at the patent-right level.

STRICT RULE: NEVER USE "" IN TEXT FIELDS

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STRICT RULE: NEVER USE "" IN TEXT FIELDS

This prompt produces only the validity dataset. Do not produce an infringement dataset or infringement rows.

This is a right-level dataset, not a claim-level dataset.

* Each row corresponds to one disputed patent or patent application, aggregated at patent level.

* Do not create separate rows merely for individual claims.

* Claim-level reasoning must be summarized inside fields 18, 25, 27, and 29.

ABSOLUTE OUTPUT RESTRICTIONS

* Output exactly one fenced code block labelled csv.

* Put all CSV rows inside that csv code block.

* Do not include a header.

* Do not include notes, explanations, citations, footnotes, source references, or any text before or after the csv code block.

* Every row must contain exactly 32 fields.

* Wrap every field in double quotes.

* Escape all internal double quotes using double double-quotes.

* Lists must look like "[""A"", ""B""]" when placed inside a CSV field.

* Dictionaries must use double quotes for all internal keys and strings, escaped correctly for CSV.

* No field may contain a URL unless that field explicitly asks for a URL.

* If the case must be skipped, output exactly one CSV row. Put "skip" in every field except field 30, which must contain the judgment URL.

* Do not hallucinate.

* If a value is not available from the judgment or required public research, leave it as "".

* Date format must be YYYY-MM-DD.

* If only month and year are available, use YYYY-MM.

* If only year is available, use YYYY.

* If a date-list field has no available information, output "[]", not "".

* Field 30 must contain the raw judgment URL only.

* Every CSV record must occupy exactly one physical line in the output.

* Never insert an actual newline or carriage-return character inside any CSV field.

* Whenever this prompt specifies \n, output the two literal characters backslash and n, not an actual line break.

* Field 26 must separate passages using the literal text \n—\n.

* In field 29, place the required patent-level outcome first, followed by the literal text \n and then the remainder of the summary.

SCOPE FILTER — ONLY THREE CASE

* be careful do not make double quote """

* link of patent must not be empty. same as all field of citations. if GB versions not available, find WO version

Extract a row only if the case belongs to one of these three types:

1. An infringement action in which the court actually considers or decides patent validity. Extract only the validity result. Do not extract infringement.

2. A revocation action or counterclaim in which patent validity is considered or decided.

3. A patent-application case in which the court considers patentability or another validity-type issue, including novelty, inventive step/obviousness, insufficiency, added matter, excluded subject matter/ non-patentability, industrial applicability, priority entitlement, or another ground bearing on whether the application may proceed to grant.

An express concession that the whole patent is invalid counts as a validity outcome under Type 1 or Type 2 even if the court does not conduct a full merits analysis of individual validity grounds. Include the patent at right level. A mere withdrawal, abandonment, non-maintenance, or agreed revocation without an express concession of invalidity does not count by itself.

For Type 3 patent-application cases, the Comptroller/UK IPO is mandatory but not necessarily the only defendant/respondent or adverse party:

* Field 6 must include "Comptroller-General of Patents, Designs and Trade Marks" and must also include every separately represented third-party respondent/opponent identified in the case.

* Field 8 must include "Comptroller-General of Patents, Designs and Trade Marks" and must also include every third-party opponent that directly opposes or challenges the patent application in this row.

* Field 10 must contain the Comptroller object and separate verified objects for all qualifying third-party opponents in field 8, subject to the historical parent/group collapse rules.

* Do not omit a qualifying third-party opponent merely because the Comptroller is already present.

* Do not add a third party merely because it appears in the caption; it must directly oppose or challenge the application.

* Field 28 = 0 when the applicant is the claimant/appellant.

Skip:

* a pure infringement or non-infringement case with no live validity issue considered, decided, or expressly conceded invalid;

* registered-design cases;

* damages, inquiry as to damages, costs-only, consequential-order, enforcement-only, or other post-merits judgments;

* a merely interlocutory judgment that does not finally or summarily decide a live validity, revocation, refusal, or patentability issue;

* permission-to-appeal decisions following a non-merits interlocutory ruling;

* any strike-out, dismissal, stay, jurisdiction, service, forum, standing, party, pleading-amendment, patent-amendment-procedure, costs, or case-management ruling that does not directly and summarily decide patent validity, revocation, refusal, or patentability;

* a row where the revocation claim, counterclaim, or only live invalidity argument is withdrawn, abandoned, not maintained, or no longer pursued and the court consequently does not decide it;

* any case outside Types 1–3 above.

SUMMARY JUDGMENT / STRIKE-OUT

Apply this section before extracting any row. First identify (a) who made the strike-out or summary-disposition application, (b) who won that application, (c) whether the application directly concerned patent validity, revocation, refusal, or patentability, and (d) whether a later hearing or judgment in the same case decided validity.

DIRECT-VALIDITY REQUIREMENT

* The strike-out or summary-disposition application must directly seek a final or summary determination of a live patent-validity, revocation, refusal, or patentability issue.

* A threshold merits assessment may count even though it is shorter than a full trial, but the court must assess the validity challenge itself rather than merely decide a collateral procedural question.

* Always skip a strike-out or other application concerning only jurisdiction, service, forum conveniens, a stay, standing, joinder or identity of parties, duplication or utility of claims, permission to amend pleadings, permission or procedure for patent amendment without deciding validity, costs, timetabling, disclosure, evidence, or other case management.

MANDATORY STRIKE-OUT DECISION MATRIX

* Challenger applies and challenger wins: INCLUDE. Treat the result as an adverse-side right-level win; field 13 = 0 and field 25 = 1. This applies when the ruling summarily disposes of the patent or the live validity issue against the owner.

* Challenger applies and patent owner wins: INCLUDE only if no later hearing or judgment in the same case decides the same patent's validity. Treat the result as an owner-side right-level win; field 13 = 0 and field 25 = 0. If a later validity hearing or judgment exists, SKIP this preliminary row and use the later validity decision instead.

* Patent owner applies and patent owner wins: INCLUDE only if no later hearing or judgment in the same case decides the same patent's validity. Treat the result as an owner-side right-level win; field 13 = 0 and field 25 = 0. If a later validity hearing or judgment exists, SKIP this preliminary row and use the later validity decision instead.

* Patent owner applies and challenger wins: SKIP. A refusal to strike out the challenger's validity case merely allows that case to continue and is not itself a patent-validity outcome.

* If the identity of the applicant, the winner, the direct validity issue, or the existence of a later same-case validity decision cannot be established reliably, do not guess. SKIP.

WITHDRAWN OR ABANDONED REVOCATION ARGUMENTS

* If a party withdraws, abandons, does not maintain, or no longer pursues a revocation claim, revocation counterclaim, or validity argument and the court consequently does not assess or decide it, do not treat the withdrawal as a validity outcome.

* If no other live validity, revocation, refusal, or patentability issue is finally or summarily decided for that patent, SKIP the entire right-level row.

* If only one ground is withdrawn but one or more other grounds are actually decided, keep the row, leave

the withdrawn ground blank in field 27, and code only the grounds the court decides.

* A formal statement that a patent is valid, maintained, or not revoked does not override this rule when it merely records the consequence of an unopposed withdrawal or amendment and the court does not assess any live validity ground.

OUTPUT FOR INCLUDED PRE-TRIAL RULINGS

* Output one right-level row per disputed patent or patent application.

* Fill all right-level fields that can be established and make field 29 state that the result is a summary or strike-out outcome rather than a full merits judgment.

* Do not invent claim-level rows or validity grounds that the court did not assess.

ROW UNIT RULE

Each row must correspond to:

* one disputed patent or patent application;

* one owner/applicant/enforcement side; and

* one adverse validity challenger side.

Do not split rows merely because:

* different claims of the same patent have different outcomes;

* the same side contains multiple co-owners, co-applicants, exclusive licensees, enforcement-side parties, or joint validity challengers; or

* the same patent is attacked on multiple validity grounds.

Split rows only when the legal/economic unit changes: a different patent/application, a different owner/applicant/enforcement side, or a different adverse validity challenger side.

If different claims of the same patent have different outcomes, aggregate at patent level. For field 25, count only claims or claim sets that were actually attacked and substantively decided. Unattacked claims, unargued claims, withdrawn or abandoned attacks, and unresolved or undecided claims do not count as surviving claims for field 25. Apply the mandatory attacked-claim and amendment rules under field 25 below.

WHOLE-PATENT INVALIDITY CONCESSION — RIGHT-LEVEL RULE

* If the judgment, final order, or reliable litigation record expressly states that the patent owner concedes that the whole patent is invalid, include one right-level row for that patent.

* A whole-patent invalidity concession is a valid adverse validity outcome even if no full merits analysis or ground-by-ground decision follows.

* Field 25 must be 1 because the patent owner loses the whole patent.

* Identify the exact historical patent/version covered by the concession.

* Field 18 must state "patent as a whole; whole patent conceded invalid" and list the independent claims of the conceded version if they can be verified.

* Field 27 must record the identified grounds. If no ground is identified, set "Others" = 1 and leave the named grounds blank.

* Field 29 must state that invalidity was conceded, identify when the concession occurred, and distinguish it from a judicial merits finding.

* An express whole-patent invalidity concession recorded before the substantive validity trial still counts.

* Mere withdrawal of a revocation application, abandonment of a patent, non-maintenance of claims, or agreed revocation without an express admission/concession that the patent is invalid does not count and must not be coded as a right-level validity loss.

PATENT-VERSION RULE — MANDATORY

The extracted patent/application must be the version legally operative and actually addressed by the judgment on the judgment date.

* Identify whether the judgment concerns the application claims, granted claims, a previously amended specification, or proposed amended claims.

* Use the correct patent/application number and publication/grant suffix applicable at that date, for example A, A1, B, or B1 where relevant.

* If the judgment does not identify the version clearly, research the official UKIPO/EPO record and contemporaneous patent documents.

* Do not silently use the current Google Patents version when it differs from the version before the court.

* Field 15 must name the correct version.

* Field 18 must identify the actual claim version attacked or decided.

* Field 29 must explain any old/granted/amended/proposed-amended distinction and identify which version was operative.

* If the court substantively assesses both the granted and amended versions, record both in fields 18 and 29 but retain one right-level row.

* If the correct historical version cannot be verified, do not guess; state the uncertainty briefly in field 29.

REFERENCE DATES

* FIRST_HEARING_DATE = the earliest hearing date in field 2 for the judgment being extracted.

* FIRST_JUDGMENT_DATE = the earliest judgment date in field 1 for the judgment being extracted.

* ENTITY_REFERENCE_DATE for fields 9 and 10 = FIRST_JUDGMENT_DATE.

* The four forward-citation counts use only FIRST_HEARING_DATE and FIRST_JUDGMENT_DATE as specified in fields 19, 20, 31, and 32.

FIELDS

1. "date_of_judgment"

List of judgment date(s), in YYYY-MM-DD format.

Example: ["2024-07-15"].

2. "date_of_hearing"

List of hearing date(s), in YYYY-MM-DD format.

If unavailable, output [].

Do not use ranges. Expand ranges into separate dates where possible.

3. "date_filing_the_complaint"

List of claim-form, revocation-application, counterclaim, or appeal filing date(s).

If unavailable, output [].

If only month and year are available, use YYYY-MM.

If only year is available, use YYYY.

4. "judges"

Names of all judges. Return a list of strings.

Do not include prefixes such as Sir, Mr, Mrs, The Hon, or Lord Justice.

If the judgment uses only a judicial title, identify the actual name if reliable public research permits.

5. "claimants"

Names of all claimants/appellants in the judgment caption or case title. Return a list of strings.

6. "defendants"

Names of all defendants/respondents in the judgment caption or case title. Return a list of strings.

For every Type 3 patent-application case, the list must include "Comptroller-General of Patents, Designs and Trade Marks". Also include every other defendant/respondent that separately opposes the application. The Comptroller is mandatory, not exclusive.

FIELDS 7–10: LEGAL PARTIES AND ECONOMIC ENTITY REPRESENTATIVES

Fields 5 and 6 are caption-level fields.

Fields 7 and 8 are row-level legal-party fields.

Fields 9 and 10 are row-level economic-entity fields measured at ENTITY_REFERENCE_DATE.

Fields 7–10 must not mechanically follow claimant/defendant status.

* Field 7 identifies the owner/applicant/enforcement legal side for the patent/application in field 15.

* Field 8 identifies the adverse validity challenger legal side.

* Fields 9 and 10 may collapse legal subsidiaries or related parties in fields 7 and 8 to their common parent/group representative under the mandatory parent rules below.

* Therefore, the standardized parent name in field 9 or 10 need not be identical to the legal-party name in field 7 or 8.

7. "right_owner_or_applicant_side"

List only the legal parties that own, apply for, exclusively license, or directly enforce the patent/application in field 15.

Rules:

* This is a patent-right owner/applicant-side field, not a claimant/defendant field.

* Include only parties connected to the specific patent/application in this row.

* Do not include parties merely because they appear in the caption.

* Do not include parties active only on another patent, another issue, damages, costs, or a formal/nominal role.

* If aligned co-owners, co-applicants, exclusive licensees, or enforcement-side parties are treated together, list them together.

* If the owner/applicant side is a defendant, still list it here.

* If ownership or applicant status cannot be determined, output [].

8. "other_side_or_validity_challenger_side"

List only the legal parties that attack, oppose, seek revocation of, or directly challenge the validity/patentability of the patent/application in field 15.

Rules:

* Include only parties connected to the specific patent/application in this row.

* Do not include parties merely because they appear in the caption.

* If several parties jointly challenge the same right and the court treats them together, list them together.

* For every Type 3 patent-application case, include "Comptroller-General of Patents, Designs and Trade Marks" and every third-party opponent that directly opposes or challenges the application in this row.

* The Comptroller is mandatory but does not replace or absorb a separate private or public opponent.

* If the adverse side cannot be determined, output [].

9. "right_owner_or_applicant_party_type_ticker_and_headquarter"

Return a dictionary indexed consecutively from "1". Each object must contain exactly these four fields and no others:

{"1":{"name":"","type":1,"ticker_at_time":[],"headquarter":0}}

10. "other_side_or_validity_challenger_party_type_ticker_and_headquarter"

Use the same JSON structure and the same mandatory verification rules as field 9.

For every Type 3 patent-application case, field 10 must contain at least:

{"1":{"name":"Comptroller-General of Patents, Designs and Trade Marks","type":3,"ticker_at_time":[], "headquarter":0}}

If field 8 contains one or more third-party opponents, add consecutively numbered objects for them. Verify and, where required, collapse each opponent under the historical entity/parent/group rules below. The Comptroller remains a separate mandatory object and must never be collapsed with another opponent.

MANDATORY FIELD 9 / FIELD 10 VERIFICATION RULES

Verify every entry as of ENTITY_REFERENCE_DATE. Do not use current ownership, listing status, ticker, or headquarters when it differs from the historical position on that date.

JSON format:

* Keys are "1", "2", and so on.

* Every object has exactly: "name", "type", "ticker_at_time", and "headquarter".

* "type" and "headquarter" must be numbers, never strings.

* "ticker_at_time" must always be a list, for example ["NYSE:ABC"]. Use [] when there was no verified ticker.

* If field 7 or field 8 is [], return {} for the corresponding field 9 or field 10.

Type coding:

* 0 = individual/person.

* 1 = private or non-listed company/entity.

* 2 = listed company. This includes a legal party that is a subsidiary of a listed parent at ENTITY_ REFERENCE_DATE; collapse it to the listed parent.

* 3 = government, UKIPO, Comptroller, or public authority.

Headquarter coding:

* 0 = United Kingdom.

* 1 = European Union excluding the United Kingdom, with Switzerland included in code 1.

* 2 = outside the United Kingdom and the code-1 region.

* For an individual/person, always use headquarter = 0 unless the judgment, the input, or reliable public research explicitly establishes a different residence or domicile.

* For the Comptroller or UKIPO, use headquarter = 0.

Parent/group representative rule:

* If a legal party was a subsidiary of a listed parent at ENTITY_REFERENCE_DATE, collapse it to the listed parent and use the parent's legal name, historical ticker, and headquarters.

* If multiple legal entities on the same side belonged to the same corporate group or ultimate parent at ENTITY_REFERENCE_DATE, collapse them into one entry.

* A UK subsidiary of Pfizer is represented by Pfizer Inc., not by the UK subsidiary.

* A German subsidiary of Clariant is represented by Clariant AG, not by the German subsidiary.

* If the ultimate parent/group was private, collapse to the private parent/group when the relationship is clear.

* If both parent and subsidiary appear on the same side, normally retain only the parent representative after collapse.

* Do not collapse entities merely because their names are similar. Require reliable evidence of common control or parentage at ENTITY_REFERENCE_DATE.

* If the relevant parent cannot be verified, retain the legal entity separately and use the safest verifiable type/ headquarter coding. Mention "insufficient evidence to collapse" briefly in field 29.

Historical ticker rule:

* The ticker must be the ticker that existed at ENTITY_REFERENCE_DATE.

* Use an exchange prefix, for example NYSE:, NASDAQ:, LSE:, EPA:, AMS:, FWB:, SIX:, SWX:, TSE:, TYO:, KRX:, BSE:, NSE:, BIT:, or EBR:.

* If a company had not yet completed its IPO at ENTITY_REFERENCE_DATE, use type = 1 and ticker_ at_time = [].

* If a company had already been acquired or delisted before ENTITY_REFERENCE_DATE, use the acquiring/group parent at that date when the parent rule applies.

* Do not use a present-day ticker for an earlier date.

Name rule:

* Standardize the legal name without unnecessary rewriting.

* When collapsing, use the parent's legal name.

* Preserve the correct suffix, such as Inc., Ltd, plc, AG, GmbH, S.p.A., N.V., Corporation, or Limited.

11. "claimant_lawyers"

Names of all lawyers representing the claimants/appellants. Return a list of strings.

If unavailable, output [].

12. "defendant_lawyers"

Names of all lawyers representing the defendants/respondents. Return a list of strings.

If unavailable, output [].

13. "stage"

Return:

* 0 = a summary judgment, strike-out judgment, or other dispositive pre-trial ruling that is INCLUDED under the mandatory SUMMARY JUDGMENT / STRIKE-OUT decision matrix above.

* 1 = merits judgment after trial or a substantive final hearing.

Skip merely interlocutory or post-merits judgments under the scope rules above.

14. "right_type"

Always return 0.

0 = patent or patent application.

Registered designs and all other rights are out of scope.

15. "right_name"

Name, title, and full publication/grant/application number of the patent or patent application.

* Use the exact historically correct version actually addressed by the judgment.

* Include prefixes and suffixes such as GB, EP, UK, WO, A, A1, B, or B1 where applicable.

* If the judgment is unclear, verify the contemporaneous version using official UKIPO/EPO records and

patent documents.

* Do not substitute a later granted or later amended version for the version before the court.

16. "right_short_description"

A concise factual description of the invention or patent application.

Use the judgment's wording where possible.

Do not speculate.

17. "google_patent_url"

Raw Google Patents URL for the patent or patent application.

You must research this online.

If no Google Patents page exists, use the best official UKIPO/EPO patent-record URL.

If no reliable URL is found, leave "".

18. "patent_unit"

Begin with "patent as a whole" or "patent application as a whole".

Then identify:

* the claims actually attacked, defended, amended, proposed for amendment, or decided;

* whether each relevant claim set is the application, old/granted, previously amended, or proposed-amended version; and

* whether revocation/refusal applies to the whole right even though only representative claims were analysed.

Examples:

* "patent as a whole; validity attacked on granted claims 1, 3, and 5"

* "patent as a whole; granted claim 1 invalid; proposed amended claim 1 allowed"

* "patent application as a whole; application claims 1–20 refused for obviousness"

* "patent as a whole; whole patent conceded invalid; independent granted claims 1, 10, and 15"

* "patent as a whole; attacked claims not individually identified"

Do not infer that unargued claims were individually invalid merely because the whole patent was revoked. An express whole-patent invalidity concession is recorded as a right-level loss under the special rule above; listing its verified independent claims in field 18 does not create separate rows in this right-level dataset.

FORWARD-CITATION RULES FOR FIELDS 19, 20, 31, AND 32

You must research Google Patents and use only the family-level "Families Citing this family" table.

General rules for all four counts:

* Count citing families, not individual publication rows.

* Each citing family counts once per field.

* Do not use the separate publication-level "Cited by" table for these four fields.

* Do not subtract self-family or self-citation entries shown by Google Patents.

* Apply the relevant cutoff date inclusively: a date equal to the cutoff qualifies.

* If there are no qualifying citing families, return 0.

* If FIRST_HEARING_DATE is unavailable, leave fields 19 and 31 empty rather than inventing a date.

19. "forward_citations_first_hearing_priority_date"

Count families in "Families Citing this family" whose priority date is on or before FIRST_HEARING_DATE.

20. "forward_citations_first_judgment_priority_date"

Count families in "Families Citing this family" whose priority date is on or before FIRST_JUDGMENT_DATE.

21. "backward_citations_1_family_to_family"

Count only the family-to-family portion of the backward citations shown by Google Patents: the rows under "Family Cites Families" or below the "Family To Family Citations" divider.

* Count the rows in this family-to-family portion only; each displayed cited family counts once.

* Exclude every ordinary patent-citation row in the first table, meaning the rows above the "Family To Family Citations" divider or before the separate "Family Cites Families" heading.

* Do not use the forward-looking "Families Citing this family" section. The direction of the words matters: "Family Cites Families" is backward; "Families Citing this family" is forward.

* Field 21 is therefore field 22 with the first ordinary patent-citation table removed.

* If the family-to-family portion is empty, return 0 even when the first ordinary patent-citation table contains rows.

* Do not impose a first-hearing or first-judgment cutoff.

* If there are no such citations, return 0.

22. "backward_citations_2_all_rows_in_patent_citations"

FIELD 22 IS ALWAYS THE NUMBER IN PARENTHESES IMMEDIATELY BESIDE "Patent Citations": "Patent Citations (N)" means field 22 = N. Extract the parenthesized N, not any number beside another heading.

* Mandatory verification formula: field 22 = number of ordinary rows in the first patent-citation table + field 21.

* The parenthesized N beside "Patent Citations" is the value to extract for field 22. Verify it by counting all backward rows: the ordinary first-table rows plus the family-to-family rows.

* Never use the forward-looking "Cited By (N)" total for field 22. "Cited By" is not a source for either backward field.

* Count the rows in each backward table exactly once within that table. Do not subtract a row merely because a related citation appears elsewhere.

* Do not impose a first-hearing or first-judgment cutoff.

* Field 22 must be greater than or equal to field 21. The difference field 22 - field 21 must equal the number of ordinary rows in the first patent-citation table.

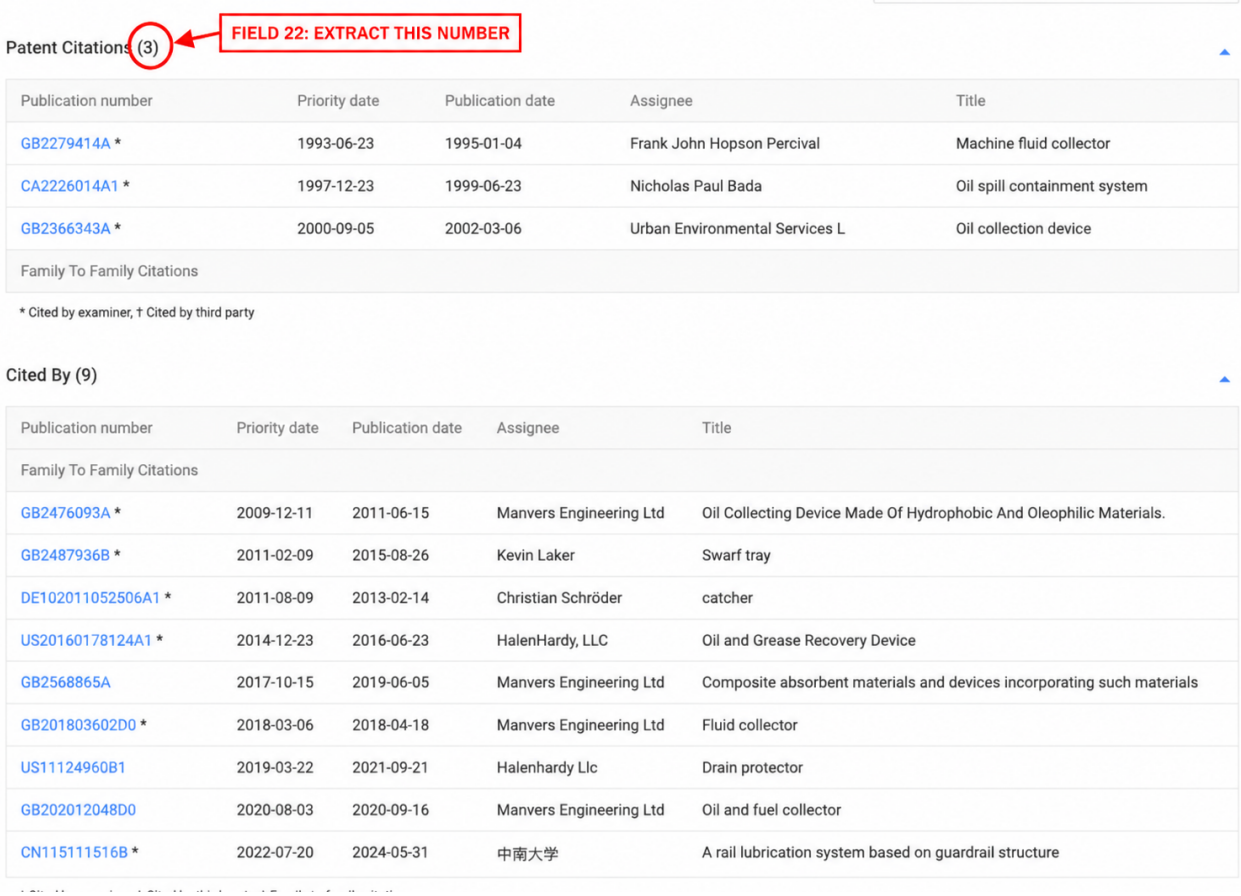
* If both the first ordinary table and the family-to-family portion are empty, return 0.

Mandatory examples:

* If the first table has 3 ordinary rows, the family-to-family portion has 0 rows, and "Cited By (9)" is shown, then field 21 = 0 and field 22 = 3. The number 9 is irrelevant.

MANDATORY VISUAL EXAMPLE — FIELDS 21–22

READ THE DIRECTIONS LITERALLY: FIELD 22 is always the parenthesized number immediately beside "Patent Citations" — "Patent Citations (N)" means field 22 = N. Backward 1 excludes the first ordinary patent-citation table; backward 2 includes the whole backward-citation block. Always verify field 22 >= field 21 and never use "Cited By (N)".



GB2428032B — the first patent-citation table has 3 ordinary backward rows; the family-to-family portion is empty; "Cited By (9)" is forward and irrelevant. Therefore field 21 = 0 and field 22 = 3.

* If the page shows "Patent Citations (61)", the first ordinary table has 1 row, and the family-to-family portion has 60 rows, then field 21 = 60 and field 22 = 61. The parenthesized 61 is field 22; any "Cited By" total is irrelevant.

23. "grant_date"

Date the patent was granted, in YYYY-MM-DD format.

For a pending or refused application that never reached grant, leave "".

24. "priority_date"

Priority date of the patent/application, in YYYY-MM-DD format.

You must research this online.

If unavailable, leave "".

25. "right_level_winner_owner_vs_other"

Code the outcome from the perspective of the patent owner or applicant:

* 0 = owner/applicant wins at right level: every attacked claim in the operative claim set survives every live validity ground asserted and decided against it; or, after amendment, every attacked claim in the final replacement amended claim set that is substantively assessed survives every live ground asserted and decided against it. A fully successful replacement amended claim set is coded 0 even if old/granted claims were deleted, abandoned, conceded invalid, or held invalid.

* 0.5 = mixed or partial outcome among attacked claims: at least one attacked claim is affirmatively upheld

and survives every live validity ground directed to that claim, but the result does not qualify for 0 because at least one other attacked claim or attacked claim set fails or remains unresolved. A claim that wins one ground but loses another, or has a material live attack left undecided, does not count as surviving.

* 1 = adverse side wins at right level: all attacked claims that are substantively decided fail, and no attacked claim is affirmatively upheld through all live grounds directed to it; this remains 1 even if other claims remain in the patent solely because they were not attacked, were not pursued, or were left undecided. Also code 1 for unconditional revocation, refusal/rejection of the application, or an express concession that the whole patent is invalid, unless a final replacement amended claim set is substantively assessed and fully survives.

MANDATORY ATTACKED-CLAIM OUTCOME RULES:

* Determine the outcome claim-by-claim first, then aggregate only across attacked claims.

* Count only claims or claim sets actually attacked on validity/patentability. A claim that was never attacked cannot rescue the owner from 1 and cannot create 0.5.

* An attacked claim counts as a survivor only if the tribunal affirmatively resolves every live validity ground that it actually decides against that claim in the owner/applicant's favour. If any decided ground succeeds against the claim, that claim is lost. If a material attack on that claim is left unresolved or undecided, do not count that claim as a survivor for field 25.

* Do not infer survival from silence. Unattacked, unargued, withdrawn, abandoned, not-maintained, or undecided claims/grounds do not create an owner win for field 25.

* A bare "revoked unless amended" order does not automatically mean 0.5. If all attacked claims decided in the judgment lose and no replacement amended claim set is substantively assessed and maintained in that judgment, code 1.

* AMENDMENT OVERRIDE: if the old/granted claims fail or are deleted, but a replacement amended claim set is then substantively assessed and every attacked amended claim survives every live ground decided against it, code 0, not 0.5. If the attacked amended set is mixed, code 0.5; if all attacked amended claims fail, code 1.

26. "product_or_process_judicial_quotes"

This field must contain raw judicial text, not a human-written summary.

Extract every materially relevant passage in which the judge describes:

* the patented invention, product, process, manufacturing method, material, component, apparatus, device, or technical field;

* a commercial embodiment; or

* a prior-art product, process, or embodiment relevant to the validity dispute.

Rules:

* Quote literally from the judgment.

* Do not paraphrase, summarize, simplify, or add words.

* Prefer fuller technical passages over generic labels.

* Put all relevant passages in the same field.

* Separate passages using the literal text \n—\n. Do not insert actual line breaks.

* Do not add ellipses unless the ellipsis appears in the judgment.

* Escape internal double quotes correctly for CSV.

* If different claim versions are in issue, include the judicial passages that identify or compare those versions. Put the human-written explanation of the version history in field 29, not here.

* Leave empty only if the judgment genuinely contains no relevant product/process/invention/prior-art description.

27. "issue_dictionary"

Return this dictionary exactly:

{

"Lack of novelty":"",

"Lack of inventive step / obviousness":"",

"Insufficiency":"",

"Added matter":"",

"Excluded subject matter / non-patentability":"",

"Priority not entitled":"",

"Lack of industrial applicability":"",

"Others":""

}

For each ground:

* 0 = patent owner/applicant wins on that ground at right level.

* 0.5 = mixed or partial outcome on that ground at right level.

* 1 = patent owner/applicant loses on that ground at right level.

* "" = the ground was not live or not decided in the present judgment.

If a ground is withdrawn, abandoned, not maintained, or not pursued and the court does not decide it, leave that ground as "". If no other ground or validity issue is decided, SKIP the entire row under the withdrawal rule above.

Code every live ground separately. Do not invent grounds.

The dictionary must be substantively consistent with field 25 but need not mechanically equal it.

For a whole-patent invalidity concession, code every expressly identified validity ground. If no specific ground is identified, set "Others" = 1 and leave all named grounds as "".

28. "claimant_is_owner_side"

Return:

* 0 if the claimant/appellant side is the patent owner or applicant side.

* 1 if the claimant/appellant side is not the patent owner or applicant side.

For revocation claims brought by a challenger, this will usually be 1.

For patent-application appeals brought by the applicant, this must be 0.

If unclear, leave "".

29. "summary"

Keep the summary under 500 words.

The first segment, before the literal text \n, must state the patent-level validity outcome clearly, such as:

* "whole patent maintained";

* "whole patent revoked";

* "whole patent conceded invalid";

* "patent partially revoked";

* "patent maintained only in amended or narrowed form";

* "revoked unless amended";

* "patent application allowed"; or

* "patent application refused".

The summary must also:

* identify the case as Type 1, 2, or 3 under the scope filter;

* identify whether the judgment follows trial/substantive final hearing or is a dispositive pre-trial ruling;

* name the patent/application and state the correct historical version considered by the judgment;

* identify the owner/applicant side and adverse validity challenger side;

* explain why the case is within scope;

* if invalidity was conceded, state expressly that the concession counts as the right-level validity outcome, identify its timing, and distinguish it from a judicial merits finding;

* identify the claims and claim versions attacked or decided without treating unargued claims as individually invalid;

* distinguish application, old/granted, previously amended, and proposed-amended claims where relevant;

* explain the result on every live validity ground;

* explain precisely why field 25 is 0.5 if the outcome is partial or mixed;

* explain why fields 7 and 8 contain the correct row-level legal parties when their alignment differs from claimant/defendant status;

* briefly state any Fields 9–10 parent/group collapse that is not obvious from fields 7–8;

* state "insufficient evidence to collapse" if a possible parent relationship could not be verified at ENTITY_REFERENCE_DATE;

* identify relevant prior art and the claims against which it was compared;

* distinguish same-case judgments from background/comparator judgments where useful;

* explain multiple complaint dates if field 3 contains more than one date; and

* contain no URLs.

30. "link"

Raw URL of the judgment only.

31. "forward_citations_first_hearing_publication_date"

Count families in the Google Patents "Families Citing this family" table whose publication date is on or before FIRST_HEARING_DATE.

Apply all general forward-citation rules stated above field 19.

32. "forward_citations_first_judgment_publication_date"

Count families in the Google Patents "Families Citing this family" table whose publication date is on or before FIRST_JUDGMENT_DATE.

Apply all general forward-citation rules stated above field 19.

FINAL VALIDATION BEFORE OUTPUT

Before returning the csv code block, verify all of the following:

1. The case belongs to exactly one of Types 1–3.

2. The output contains validity only and no infringement row.

3. Type 3 includes the Comptroller/UKIPO in fields 6, 8, and 10 and also retains every qualifying third-party opponent in those fields.

4. An express whole-patent invalidity concession is included with field 25 = 1, while withdrawal, abandonment, non-maintenance, or agreed revocation without an express invalidity concession is not misclassified.

4A. Every summary judgment or strike-out row passes the mandatory applicant/winner matrix, directly concerns validity, and has been checked for a later same-case validity hearing or judgment.

4B. If revocation or the only live validity argument was withdrawn or abandoned and the court did not decide it, the row is skipped; a withdrawn individual ground is left blank when other grounds were decided.

4C. Field 25 follows the mandatory attacked-claim rule: no attacked claim affirmatively survives all of its live grounds = 1; at least one attacked claim affirmatively survives all of its live grounds, but the attacked set is not a complete win = 0.5; all attacked claims in the operative set win = 0; and a fully successful substantively assessed replacement amended claim set = 0 even if old/granted claims were deleted or invalid.

5. The patent/application version is correct as of the judgment date and is the version actually addressed by the judgment.

6. Fields 9 and 10 use historical entity, parent, ticker, and headquarter information as of FIRST_ JUDGMENT_DATE.

7. Fields 9 and 10 use numeric type/headquarter values and list-valued ticker_at_time.

8. Forward fields 19, 20, 31, and 32 are family-to-family counts and use the correct date field and cutoff date.

9. Field 21 counts only the backward family-to-family portion and excludes every ordinary row in the first patent-citation table; it never uses the forward "Families Citing this family" section.

10. Field 22 is always the parenthesized N immediately beside "Patent Citations". Verify that N equals the number of ordinary first-table rows plus field 21; never use "Cited By (N)".

11. Field 22 is greater than or equal to field 21, and field 22 minus field 21 equals the number of rows in the first ordinary patent-citation table. If either check fails, recount before output.

12. Every row has exactly 32 double-quoted fields.

13. Every CSV record occupies exactly one physical line, with no actual newline or carriage-return character inside any field.

14. The answer contains exactly one csv code block and no text outside it.

D.2 Second prompt: claim-level validity attacks

Validity-Attack Extraction Prompt

Claim-level patent validity rows | eight attack categories | max-parent dependency_depth | prior-art unit IDs

Updated version: dependency_depth is computed by the max-parent rule. Independent claims are 0. A dependent claim has dependency_depth = 1 + the maximum dependency_depth of its direct parent claims. claim_text_expanded must be a standalone technical definition, must preserve alternative dependency paths as alternatives rather than cumulative limitations, and must contain no patent-claim cross-reference. An express concession must be classified under a specific attack category whenever the judgment identifies one; attack category 8 is only the residual category for expressly conceded invalidity with no identified ground in categories 1-7. For novelty and obviousness, each prior-art document/component is output on its own CSV row, while the final field prior_art_unit_id groups consecutive component rows that the judgment treats together as one prior-art unit. Mandatory discrepancy warnings are now a controlling exception to the eight-block-only output rule: every input claim with zero output rows must be named visibly, and every mismatch between issue_dictionary and the judgment must also be reported. Claim-set completeness is checked only within the patent or patents actually submitted in the current request: warn for missing or extra input claims within each submitted patent, but never warn about claims from other patents in the same judgment that were not submitted.

1. Inputs

You will receive three inputs:

One or more claims text files, covering either one patent from the judgment or multiple patents from the judgment.

A BAILII, Casemine, archive, or other judgment text file or judgment URL.

One or more CSV input rows.

Each CSV input row contains, in order:

claims_text_file_link

judgment_url

google_patents_url

target_claim_label

index

issue_dictionary

Example input row:

file:///Users/nghiapham/Documents/claim/text/EP0958447B1_claims.txt

https://www.bailii.org/ew/cases/EWPCC/2006/49.html

https://patents.google.com/patent/EP0958447B1/en

claim 1

10000

'{"Lack of novelty":0,"Lack of inventive step / obviousness":0,"Insufficiency":"","Added matter":"","Excluded subject matter / non-patentability":"","Priority not entitled":"","Lack of industrial applicability":"","Others":"", "Conceded invalidity":""}'

The issue_dictionary is to double check. If there is a number rather than null (“”””) it means it had this ground when you did this the first time. It is not ground truth, you might have made a mistake the first time but make sure to check it. Its possible you hallucinated things that didn't exist, it is also possible you forgot a ground of attack.

2. Task

Your task is to extract validity-attack data for the target claim identified in target_claim_label. If multiple CSV input rows are provided, process each target claim independently and preserve the first five fields of each row exactly.

3. Eight possible validity attacks

Section name | Attack type

novelty/ số 1 | Novelty

obviousness/ số 2 | Obviousness / inventive step

added matter/ số 3 | Added matter

lack of industrial applicability/ số 4 | Lack of industrial applicability

excluded subject matter/ số 5 | Excluded subject matter

extension by amendment/ số 6 | Extension by amendment

insufficiency/ số 7 | Insufficiency

concede invalid/ số 8 | Conceded invalidity - unspecified ground only

4. Output format

Output exactly eight sections, in the exact order shown above.

Under each section name, output exactly one CSV code block.

If a validity attack is actually addressed for the target claim, put the relevant CSV row or rows inside that attack's CSV code block.

If a validity attack is not actually addressed for the target claim, still output that attack section and still output a CSV code block, but leave the CSV code block completely empty.

Do not output JSON.

Do not include a CSV header.

Do not include explanation, comments, bullet points, or extra text outside the eight attack section names and the eight CSV code blocks, except for the mandatory discrepancy WARNING lines required by Section 4A.

4A. Mandatory discrepancy warnings - explicit exception to the eight-block rule

Before final output, determine the submitted-patent scope, build an expected-claim inventory for each submitted patent, and build an internal claim-coverage ledger for every CSV input row. For each input target claim, identify every validity ground actually addressed by the judgment and count all output rows across the eight CSV blocks. Check issue_dictionary against the judgment, but treat the judgment as ground truth.

Submitted-patent scope: the current request may contain all target claims for only one patent from a multi-patent judgment, or it may contain all target claims for multiple or all patents in that judgment. Treat only the patent or patents represented by the claims text file(s), Google Patents URL(s), and CSV input rows deliberately supplied in the current request as submitted. Do not assume that every patent mentioned in the judgment has been submitted merely because the judgment discusses it.

Perform the completeness check separately for each submitted patent. If the current request submits only Patent A, check missing and extra claims only for Patent A. Do not issue any missing-claim, missing-row, or missing-patent warning for Patent B, Patent C, or any other patent from the same judgment that was not submitted in the current request. If multiple patents are submitted, repeat the check independently for each one.

For each submitted patent, build expected_claim_inventory from every claim or separately treated claim version of that patent whose validity the judgment actually addresses on at least one of the eight grounds. Distinct granted, amended, proposed-amended, corrected, or other claim versions count as distinct targets only when the judgment treats them separately. Do not include claims of an unsubmitted patent in this inventory.

Compare the CSV input rows for each submitted patent against that patent's expected_claim_inventory. Match by patent identity and the complete target_claim_label, including any version qualifier such as granted, amended, proposed amended, corrected, claim-set label, or dependency qualifier. Do not match merely by bare claim number when the judgment distinguishes versions.

If an expected claim or separately treated claim version of a submitted patent has no CSV input row, output exactly one visible warning in this format: WARNING - MISSING INPUT CLAIM: patent="[patent identifier]"; expected_target_claim_label="[expected target claim label]"; the judgment addresses this claim of a patent submitted in the current request on at least one of the eight validity grounds, but no CSV input row was supplied; no CSV row can be produced for this missing input claim.

If a CSV input row does not correspond to any claim or separately treated claim version in expected_claim_ inventory for its submitted patent, output exactly one visible warning in this format: WARNING - EXTRA INPUT CLAIM: patent="[patent identifier]"; target_claim_label="[target_claim_label]"; index="[index]"; this input claim is not an expected validity target for this submitted patent in the judgment; review or remove this CSV input row.

A duplicate CSV input row for the same submitted patent and the same complete target_claim_label is also extra unless the judgment genuinely treats the rows as distinct claim versions and the labels distinguish those versions. Multiple output rows generated from one legitimate input claim because of different prior-art units or component documents do not make the input claim extra.

The missing/extra completeness check concerns the CSV input claim set, not the complete numbered claim set printed in the patent. A claim that merely exists in the claims text file is not expected unless the judgment actually addresses its validity on at least one of the eight grounds. Conversely, a claim of a submitted patent that the judgment does address on such a ground must be reported as missing if its CSV input row is absent.

If an input target claim produces zero CSV rows because the judgment does not attack that claim on any of the eight validity grounds, output exactly one visible warning for that input row in this format: WARNING - ZERO OUTPUT ROW: target_claim_label="[target_claim_label]"; index="[index]"; the judgment does not attack this target claim on any of the eight validity grounds; no CSV row is output for this input claim.

The ZERO OUTPUT ROW warning is mandatory even when issue_dictionary is entirely empty. An input target claim that produces no CSV row must never silently disappear from the final answer.

If the judgment attacks an input target claim on a ground whose issue_dictionary field is empty, output one visible warning in this format: WARNING - MISSED GROUND: target_claim_label="[target_claim_label]"; index="[index]"; issue_dictionary leaves "[attack category]" empty, but the judgment attacks this target claim on that ground; the corresponding CSV row(s) are included.

If issue_dictionary lists a ground for an input target claim but the judgment does not attack that target claim on that ground, output one visible warning in this format: WARNING - HALLUCINATED GROUND: target_claim_label="[target_claim_label]"; index="[index]"; issue_dictionary lists "[attack category]", but the judgment does not attack this target claim on that ground; no CSV row is output for that ground.

Apply the missing-input-claim, extra-input-claim, zero-output-row, missed-ground, and hallucinated-ground warning tests as specified above. Output one WARNING line for each applicable discrepancy. Place all WARNING lines in the final answer immediately before novelty/ số 1. These WARNING lines are mandatory and are the only permitted text outside the eight section names and eight CSV code blocks. This exception overrides every instruction in this prompt that says "nothing else", "no extra text", or "only the eight sections". Never hide a required warning in analysis, reasoning, commentary, a CSV block, or any content that may be collapsed.

If none of these discrepancies exists, output no WARNING line and follow the normal eight-section-only format.

5. Row granularity

For each validity attack that is actually addressed for the target claim, output the CSV row or rows required by the rules below.

For novelty and obviousness, first identify each distinct prior-art unit considered by the judgment. A prior-art unit may consist of one document, multiple documents treated together as one combined reference/disclosure/attack, or Common general knowledge. Output one CSV row per component document, not one row per prior-art unit. A one-document unit therefore produces one row; a multi-document unit produces one row for each component, with all of those component rows grouped by the same prior_art_unit_id. Exception: if the patent owner expressly concedes that the target claim lacks novelty or is obvious but the judgment identifies no prior-art unit for that concession, output exactly one row for the target claim under that specific attack and leave prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id empty.

For obviousness/inventiveness, Common general knowledge may be output as its own single-row unit if the judgment actually addresses it as part of the obviousness analysis for the target claim. Give that row its own prior_art_unit_id.

Do not combine Common general knowledge and a named prior-art document in the same CSV row. If the judgment treats them together as one combined obviousness unit, output separate component rows and assign the same prior_art_unit_id; otherwise assign different IDs.

Do not collapse genuinely distinct prior-art units together. If the judgment treats two or more documents together as one prior-art reference, one combined disclosure, or one combined attack, output each component document on its own CSV row and give every component row of that unit the same prior_art_unit_id.

Keep the component rows for one unit consecutive.

Treat multiple documents as the same prior-art unit when they are copies, translations, versions, parts, appendices, or evidentiary records of the same underlying prior-art disclosure and the judgment analyzes them as one unit. In that case, output separate component rows with the same prior_art_unit_id. Do not assign the same ID merely because documents concern the same technology, belong to the same patent family, have the same author, or are cited near each other if the judgment analyzes them as distinct prior art.

For a multi-document prior-art unit, prior_art_name_less_than_20_words must identify only the specific component document represented by that row. Do not concatenate multiple component names with " + ", semicolons, pipes, or any other delimiter. Keep each component name under 20 words.

For novelty and obviousness, assign prior_art_unit_id as a zero-based integer string in judgment order, starting at "0" separately for each target claim and each attack section. All rows belonging to the same prior-art unit must have the same ID and must be consecutive. Different prior-art units must have different IDs. Example: if documents A and B are treated together as one prior-art unit and document C is a separate unit, output A with prior_art_unit_id = "0", B immediately after it with prior_art_unit_id = "0", and C with prior_art_unit_id = "1".

Do not group different validity attacks together. prior_art_unit_id is local to the target claim and attack section and does not link rows across novelty and obviousness.

For added matter, lack of industrial applicability, excluded subject matter, extension by amendment, insufficiency, and residual concede invalid, output one CSV row for the attack if that attack is actually addressed for the target claim, unless the judgment clearly treats multiple distinct attacks of the same type separately for the same target claim.

Apply ground-specific classification before concede invalid. If the judgment states that the patent owner concedes, accepts, or no longer contests invalidity for lack of novelty, obviousness, added matter, lack of industrial applicability, excluded subject matter, extension by amendment, or insufficiency, output the row only in the corresponding specific section (1-7), with patent_owner_loses = "1". The fact that the result arose by concession does not create an additional concede invalid row.

Use concede invalid (section 8) only as a residual category: the judgment must expressly concede, accept, or no longer contest the invalidity of the target claim, and must not identify any corresponding ground in sections 1-7 for that concession. Do not infer a concession from amendment, withdrawal, silence, non-appearance, or the general result of the case. Do not infer a specific ground from an unspecified concession.

For every residual concede invalid row, patent_owner_loses must be "1" and prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id must all be empty. Classify separately claim by claim: a ground-specific concession for one target claim does not determine the classification of another target claim.

6. Output row schema

Each output row must begin with the exact same five fields from the input CSV row, in the same order. After those five original fields, append the following extra fields in this exact order:

claim_text_literal

claim_text_expanded

dependency_depth

patent_owner_loses

prior_art_name_less_than_20_words

amended_claim

prior_art_link_status

prior_art_full_text_link

prior_art_unit_id

7. Field definitions

Field | Instruction

claim_text_literal | Output the literal text of the target claim. Prefer the wording as reproduced in the judgment. If the judgment does not reproduce the full claim text, use the claims text file. Do not paraphrase. For amended_claim = "1", the controlling text is the amended, proposed-amended, corrected, or otherwise modified claim version reproduced, approved, or analyzed in the judgment, including any schedule, annex, or order. Use the claims text file only as a fallback for wording that the judgment does not reproduce and only after confirming that the relevant wording was not amended, deleted, renumbered, disclaimed, or replaced. Never substitute superseded wording from the old claim set.

claim_text_expanded | For amended_claim = "1", first reconstruct the controlling amended claim set from the judgment, including all renumbering, changed dependency references, inserted limitations, deletions, substitutions, provisos, and disclaimers. Build claim_text_expanded from that amended claim set and use the judgment's amended wording wherever it is available. Do not trace a direct parent or ancestor by the numbering or dependency graph of a superseded claims text file. The claims text file is only a fallback for the exact wording of a portion that the judgment does not reproduce and that is demonstrably unchanged. Never mix wording or dependency nodes from different claim versions in one expansion. If dependency_depth = "0", this must be exactly identical to claim_text_literal. If dependency_depth is greater than "0", construct a standalone technical definition that preserves every substantive technical limitation of the target claim and every inherited limitation required by each applicable dependency path. First identify the target claim's direct parents from its dependency wording. Recursively expand each direct parent along its own path until an independent claim is reached; do not stop at the immediate parent. Remove or minimally rewrite only patent-claim cross-references and dependency scaffolding such as "Claim 1:", "according to claim 4", "claims 2-4", or "the solution of claim 5". Preserve the substantive wording literally as far as possible and add only the minimum bridging language needed to produce grammatical standalone text. Do not remove the word "claim" when it is part of the substantive technical subject matter, such as an insurance-claim processing system. For alternative direct parents p1, p2, ..., first construct one complete expansion for each path. The result must be logically equivalent to [E(p1) + N(c)] OR [E(p2) + N(c)] OR ..., where E(p) is the recursively expanded parent text and N(c) contains the substantive limitations newly added by the target claim. Present these branches in one field using "or", "any one of the following alternatives", or equivalent standalone wording. Identical wording shared by every branch may be factored out only when the result remains exactly logically equivalent. Never combine branch-specific limitations cumulatively. Every limitation newly added by the target claim must apply to every alternative. Neutral bridging language such as "no additional limitation as to dosage form" may be introduced only when needed to describe a branch that adds no limitation on that dimension; it must not add new technical substance.

dependency_depth | Output the dependency depth using the max-parent rule: independent claim = "0"; dependent claim = 1 + the maximum dependency_depth of all direct parents selected by the target claim's dependency wording. Claims inherited only through a direct parent are ancestors and enter recursively; do not add them again as direct parents. Use "1", "2", "3", etc. Do not cap at "2" unless explicitly instructed. See Section 8 for examples. For amended_claim = "1", identify every direct parent and calculate every depth from the controlling amended claim set, never from the old claims file.

patent_owner_loses | Output "0" if the patent owner wins on this validity attack for this target claim. Output "1" if the patent owner loses. For novelty and obviousness, decide the outcome separately for each prior-art unit or Common general knowledge unit. If one prior-art unit has multiple component rows, repeat the same patent_owner_loses value on every row sharing that prior_art_unit_id because those rows represent one combined attack. A ground-specific concession and a residual concede invalid row both require "1" because the patent owner has expressly conceded, accepted, or ceased to contest invalidity of the target claim on the relevant basis. Do not use outcomes from other claims. Do not guess from the general result of the case.

prior_art_name_less_than_20_words | For novelty and obviousness, output the name of the specific prior-art document/component represented by this row, or "Common general knowledge" for a Common

general knowledge row. If multiple documents form one prior-art unit, do not combine their names in one field; each component must have its own row and its own concise name under 20 words. The shared prior_art_unit_id, not name concatenation, indicates that multiple rows belong to one combined prior-art unit. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field. For all other validity attacks, output an empty CSV field.

amended_claim | Output "1" if the target claim is an amended claim. Output "0" if the target claim is not an amended claim.

prior_art_link_status | For novelty and obviousness only, assess this field separately for the specific component document represented by the row. Output "2" if prior_art_full_text_link provides full text for that component; output "1" if it provides only an abstract, summary, extract, citation record, bibliographic record, citation in another document, or another identifiable non-full-text description; output "0" if no identifiable source link is available for that component. For a multi-document prior-art unit, do not calculate one aggregate link status for the whole unit; each component row has its own status. For Common general knowledge, output "0". If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field rather than "0". For all other validity attacks, output an empty CSV field.

prior_art_full_text_link | For novelty and obviousness, output the single best available link for the specific prior-art document/component represented by this row. Prefer full text. For patent prior art, prefer a Google Patents full-text link if the patent number is clearly identified. For non-patent prior art, use the best available reliable link if full text is unavailable, including an abstract, summary, bibliographic record, citation record, or another supplied document that clearly identifies the same specific reference. Never place links for two or more prior-art components in one CSV field, whether separated by semicolons, " | ", commas, "+", or any other delimiter. If multiple documents form one prior-art unit, output separate component rows and group them with the same prior_art_unit_id. For Common general knowledge and all non-novelty/non-obviousness attacks, output an empty CSV field. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, also output an empty CSV field.

prior_art_unit_id | For novelty and obviousness only, use this final field to identify which component rows belong to the same prior-art unit. For each target claim and each attack section separately, number distinct prior-art units in judgment order using zero-based integer strings: "0", "1", "2", and so on. All component rows of one multi-document unit must share the same ID and be consecutive. A one-document unit has one row with its own ID. Common general knowledge, when output as its own unit, also receives its own ID. Do not reuse an ID for a different unit within the same target claim and attack section. Numbering may restart at "0" for a different target claim or for the other attack section. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field. For added matter, lack of industrial applicability, excluded subject matter, extension by amendment, insufficiency, and residual concede invalid, output an empty CSV field.

8. Dependency-depth rules

dependency_depth = "0": target claim is independent; claim_text_expanded must be identical to claim_text_literal.

Mandatory source-of-truth rule for amended claims: when amended_claim = "1", treat the amended or proposed-amended claim set stated in the judgment, schedule, annex, or order as the controlling and closed dependency graph. Every claim number in a dependency reference means the claim with that number in that amended set. NEVER trace the target, a direct parent, or an ancestor through the numbering or dependency wording of the old or granted claims file.

For an amended claim, use the judgment's wording for the target claim and for every amended parent or ancestor wherever that wording is provided. The claims text file may supply only exact wording omitted from the judgment and confirmed to be unchanged. If the judgment changes claim 1 but leaves a dependent claim's own wording unchanged, expand that dependent claim through the amended claim 1, not the old claim 1.

Apply every amendment that changes substantive scope, even if it appears outside the numbered claim text: include inserted disclaimers, provisos, substitutions, deletions, and limitations stated in the judgment, schedule, annex, or order. Do not omit such text merely because it is absent from the claims text file.

Renumbering/version check: before expanding an amended target, make an internal mapping of each amended claim number to its direct parent or parents. If the judgment omits, deletes, consolidates, or renumbers old claims, do not carry the old numbering forward and do not assume that amended claim N corresponds to old claim N.

Mandatory trace check for amended claims with dependency_depth of "2" or greater: write an internal path ledger for every alternative path, listing the target, each direct parent, every ancestor, the claim-set version/ source used for that node, and the calculated depth. Verify that every path ends at an independent claim in the same amended claim set and that no node was silently taken from a superseded claim set. If the amended dependency path cannot be reconstructed unambiguously from the available materials, do not guess or import the old path.

For any dependent claim, first identify its direct parents from the dependency wording in the applicable controlling claim set. A direct parent is a claim expressly selected by the dependency phrase. An earlier claim inherited only through a direct parent is an ancestor, not an additional direct parent. Calculate recursively from that same controlling claim set: dependency_depth(independent claim) = 0; dependency_ depth(dependent claim) = 1 + max(dependency_depth(direct parent) for every direct parent selected by the target claim).

If a claim refers to "any preceding claim", "any one of the preceding claims", "any of claims 1 to N", or similar wording, treat every claim in that referenced range as a direct-parent alternative for the target claim. Ancestors of those alternatives enter only through recursive expansion.

dependency_depth = "1": every expressly listed parent claim is independent, e.g. claim 3 according to claim 1 or claim 2 where both claim 1 and claim 2 are independent.

dependency_depth = "2": at least one expressly listed parent has dependency_depth "1" and no expressly listed parent has a greater depth, e.g. claim 3 according to claim 1 or claim 2 where claim 1 is independent and claim 2 depends on claim 1.

Use "3", "4", etc. for deeper dependency chains. Do not cap at "2" unless explicitly instructed.

Do not use the shortest path where alternative direct parents have different depths. Use the maximum direct-parent depth because claim_text_expanded must account for every dependency alternative selected by the target claim.

Do not create extra rows merely because a dependent claim has alternative dependency paths, unless the judgment actually treats those alternatives differently for validity. Preserve the "or" dependency language in one expanded text where appropriate.

When expanding a dependent target with alternative direct parents, construct each complete path separately before combining the paths with "or". Write claim_text_expanded as one standalone technical definition using wording such as "any one of the following alternatives: ...; ...; or ...". Do not label branches as patent claims and do not leave patent-claim cross-references in the expanded text.

If preserving literal dependency wording would leave a patent-claim cross-reference inside claim_text_ expanded, rewrite only the dependency-reference wording into standalone language. Preserve every substantive technical limitation. The first priority is logical equivalence: no missing inherited limitation, no new technical limitation, and no misleading cumulative combination of alternative branches.

Alternative-path check: if the direct parents are p1, p2, ..., verify that the expanded text means [E(p1) + N(c)] OR [E(p2) + N(c)] OR ..., not E(p1) + E(p2) + ... + N(c).

Worked example: claim 1 is independent; claim 2 depends on claim 1; claims 3 and 4 each depend on claim 2; and claim 5 depends on any one of claims 2-4 while adding a concentration limitation. The direct parents of claim 5 are claims 2, 3, and 4. Its longest path is 5 -> 3 or 4 -> 2 -> 1, so dependency_depth = "3".

For the worked example, the three alternatives are: the composition inherited through claim 2 with no additional dosage-form limitation; the same composition inherited through claim 3 and restricted to a cream; or the same composition inherited through claim 4 and restricted to a gel. The concentration limitation added by claim 5 applies to all three alternatives.

A correct claim_text_expanded for the worked example is: "A pharmaceutical composition containing calcipotriol monohydrate: (1α, 3β, 5Z, 7E, 22E, 24S)-24-cyclopropyl-9,10-secochola-5,7,10(19),22-tetraene-1, 3,24-triol, monohydrate, and having an active-component content of 1-100 μg/g of the composition, where either (i) no further limitation is imposed on the dosage form; (ii) the composition is a cream; or (iii) the composition is a gel."

9. Prior-art link rules

Do not leave prior_art_full_text_link empty merely because the available source is only an abstract, summary, extract, citation record, bibliographic record, or citation in another document.

The link must identify the same specific prior-art reference being compared.

Do not output an unrelated modern page, generic search result, or source that does not clearly identify the prior-art reference.

If a prior-art component is identified but no identifiable link is available for that component, output an empty prior_art_full_text_link and set that row's prior_art_link_status to "0". If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit at all, leave prior_art_name_less_ than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id empty.

When multiple documents form one prior-art unit, output one consecutive CSV row per component document. Put only that component's best available link in prior_art_full_text_link. Never put multiple component links side by side in one field. Give all component rows of the same unit the same prior_art_unit_id.

For a multi-document prior-art unit, determine prior_art_link_status separately on each component row: "2" if full text is available for that component, "1" if only an identifiable non-full-text source is available for that component, and "0" if no identifiable source link is available for that component. Do not downgrade or upgrade one component because of link availability for another component.

10. Important rules

Always output all eight attack sections and all eight CSV code blocks.

Do not omit an attack section merely because the attack is not addressed.

If an attack is not addressed for the target claim, its CSV code block must be empty.

Do not infer an attack merely because it is mentioned elsewhere in the judgment.

The validity attack must be applied to the target claim.

A concession must first be assigned to any specific validity ground expressly identified by the judgment. For example, a claim conceded to be obvious belongs only in obviousness/ sṓ 2, with patent_owner_loses = "1"; do not duplicate it in concede invalid/ sṓ 8. Apply the same rule to every specific ground in sections 1-7.

A concede invalid row requires an express concession or acceptance of invalidity as to the target claim and is permitted only when no specific ground in sections 1-7 is identified for that concession. Do not infer it from a concession concerning another claim, from a proposed amendment, or from the patent-level outcome.

For residual concede invalid, patent_owner_loses must be "1". The prior-art name, link-status, link, and prior_art_unit_id fields must be empty.

Preserve the first five input fields exactly as CSV fields.

Quote every CSV field with double quotes.

Escape internal double quotes by doubling them.

Do not put line breaks inside any CSV field. Each CSV row must be one single physical line.

For an empty CSV field, output "".

The prior_art_link_status field must appear immediately before prior_art_full_text_link, and prior_art_ unit_id must appear immediately after prior_art_full_text_link as the final field of every output row.

For novelty and obviousness rows tied to an identified prior-art unit, prior_art_link_status must be "2", "1", or "0" for the specific component row and prior_art_unit_id must be a zero-based integer string identifying

its unit. For a ground-specific concession of lack of novelty or obviousness with no identified prior-art unit, prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id must all be empty. For all other validity attacks, prior_art_link_status and prior_art_unit_id must be empty CSV fields.

Do not output null, None, N/A, unknown, or any other placeholder.

Be careful not to output malformed CSV such as quadruple quotes.

The final answer must contain exactly eight section names and exactly eight CSV code blocks. The only permitted additional text is the mandatory discrepancy WARNING lines required by Section 4A.

11. Mandatory internal validation before final output

Create the submitted-patent scope, per-patent expected_claim_inventory, and claim-coverage ledger required by Section 4A before drafting the final answer. The ledger must contain one entry for every CSV input row, keyed by patent identity, target_claim_label, and index.

For each submitted patent, compare expected_claim_inventory against the CSV input rows for that patent. Emit one MISSING INPUT CLAIM warning for every expected claim or separately treated claim version with no input row, and one EXTRA INPUT CLAIM warning for every supplied row that does not correspond to the expected inventory or is an unjustified duplicate.

Recheck the patent-scope boundary before emitting completeness warnings. Never emit a missing warning for a claim belonging only to a patent from the same judgment that was not submitted in the current request.

For every input row, count its CSV rows across all eight attack blocks. If the count is zero, verify that the final answer contains the mandatory ZERO OUTPUT ROW warning for that input row, even when issue_dictionary is entirely empty.

Compare the judgment-determined attack grounds with issue_dictionary for every input target claim. Emit a MISSED GROUND warning for every judgment-only ground and a HALLUCINATED GROUND warning for every issue_dictionary-only ground.

Do not finalize while any submitted patent's expected claim is missing without a MISSING INPUT CLAIM warning, any extra input claim lacks an EXTRA INPUT CLAIM warning, any input target claim has silently disappeared, or any other required discrepancy warning is absent. Required WARNING lines must be visible immediately before novelty/ sṓ 1 in the final answer, not merely mentioned in analysis or commentary.

This validation checklist is internal only. Do not output the checklist. Output any mandatory discrepancy WARNING lines required by Section 4A, followed by exactly the required eight sections and eight CSV code blocks.

Reconstruct the dependency chain for every target claim before drafting rows. For an unamended claim, use the applicable claims text. For amended_claim = "1", use the controlling amended claim set from the judgment, schedule, annex, or order; use the claims text file only for wording that is omitted and confirmed unchanged.

For every amended_claim = "1" row, verify that the target, every direct parent, and every ancestor belong to the same controlling amended claim set. Fail the row if any dependency node or claim number was taken from the old or granted claim set merely because it was available in the claims text file.

For every amended claim with dependency_depth of "2" or greater, inspect the complete internal path ledger node by node and branch by branch. Verify the amended numbering, direct-parent choices, maximum depth, independent endpoint, and source/version for every node before accepting claim_text_expanded.

For every amended claim, compare claim_text_literal and claim_text_expanded against the judgment's amendment language and verify that all inserted limitations, disclaimers, provisos, substitutions, deletions, and renumbering effects are reflected. Do not allow old wording to overwrite or dilute amended wording.

If dependency_depth = "0", verify claim_text_expanded is exactly identical to claim_text_literal.

If dependency_depth > "0", verify claim_text_expanded is not identical to claim_text_literal.

If dependency_depth > "0", verify claim_text_expanded preserves every substantive technical limitation of

the target dependent claim and incorporates every inherited limitation required by each applicable dependency path. Patent-claim cross-references and dependency scaffolding may and must be removed or minimally rewritten.

If dependency_depth > "0", verify claim_text_expanded contains no patent-claim cross-reference or dependency scaffolding such as "claim 1", "claims 2-4", or "according to claim 3". Do not reject a row merely because the word "claim" forms part of the substantive technical subject matter.

If the target has alternative dependency paths, verify claim_text_expanded preserves the alternative logic using “or”, “any one of the following”, or equivalent standalone wording, rather than stacking all parent alternatives cumulatively.

For multi-level dependencies, do not stop at the immediate parent claim. Continue until an independent claim is reached.

Verify the numeric dependency_depth follows the recursive max-parent rule: independent claims are 0; dependent claims equal 1 plus the maximum dependency_depth of every direct parent selected by the dependency wording.

If dependency_depth and claim_text_expanded are inconsistent, the row is invalid and must be fixed before output.

Verify amended_claim is correct for the target claim. If amended_claim = "1", also verify that claim_text_literal, claim_text_expanded, dependency_depth, and every dependency path were derived from the same controlling amended claim set rather than the old claims file.

Verify patent_owner_loses is decided separately for each attack and each prior-art unit. If a unit has multiple component rows, verify every row sharing its prior_art_unit_id repeats the same patent_owner_loses value.

For novelty and obviousness, verify that every specific prior-art document/component has its own CSV row. If multiple documents are treated together as one prior-art unit, verify that their rows are consecutive and share the same prior_art_unit_id. Verify that genuinely distinct prior-art units have different IDs.

For every multi-document prior-art unit, verify that no component names or links are concatenated in one field; each row contains only one component name, one component-specific prior_art_link_status, and at most one best link. Verify that all component rows share the same zero-based prior_art_unit_id, are consecutive, and that IDs increase in judgment order starting at "0" for each target claim and attack section.

For every express concession, verify first whether the judgment identifies a specific ground in sections 1-7. If it does, verify that the row appears only in that specific section, patent_owner_loses is "1", and no duplicate row appears in concede invalid/ số 8.

For a ground-specific concession of lack of novelty or obviousness with no identified prior-art unit, verify that exactly one row is output for the target claim in the specific section and that prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id are all empty; do not invent a prior-art name, unit ID, or use link status "0".

For every concede invalid row, verify that the concession is express, applies to the target claim, identifies no ground in sections 1-7, and has patent_owner_loses = "1".

For every residual concede invalid row, verify that prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id are empty.

Verify no CSV row contains line breaks, every field is double-quoted, internal double quotes are escaped by doubling them, and no malformed quadruple quotes appear.

12. Required final answer skeleton

[zero or more mandatory WARNING lines under Section 4A, only when applicable]

novelty/ số 1

```csv
```

obviousness/ số 2

```csv
```

added matter/ số 3

```csv
```

lack of industrial applicability/ số 4

```csv
```

excluded subject matter/ số 5

```csv
```

extension by amendment/ số 6

```csv
```

insufficiency/ số 7

```csv
```

concede invalid/ số 8

```csv
```

Final check: make sure claim_expanded is the same for the same claims

Final check: with the exception of novelty and obviousness, each claim only has 1 row for other fields.

Final check: if the judgment copies the exact prior art then it is also type 2

Final check: same prior arts must have the same link. For example link for prior art A used for claim 1 must have the same link as prior art A used for claim 2

Final check: verify the submitted-patent scope, each submitted patent's expected_claim_inventory, and the claim-coverage ledger. Every expected validity target of every submitted patent must have an input row or a visible MISSING INPUT CLAIM warning; every extra input claim must have an EXTRA INPUT CLAIM warning. Do not check or warn about missing claims from patents in the judgment that were not submitted in the current request. Every input target claim must either produce at least one CSV row or a visible ZERO OUTPUT ROW warning. Every ground mismatch between issue_dictionary and the judgment must produce a visible MISSED GROUND or HALLUCINATED GROUND warning. Place these warnings immediately before novelty/ số 1; this requirement overrides every "nothing else" rule in this prompt.

D.3 Third prompt: patent text extraction

Patent extraction

attention: before extracting make sure this is the correct patent in the case

You are given a Google Patents link or a patent PDF. Extract only the English patent text and create exactly two downloadable UTF-8 plain text files.

IMPORTANT SOURCE RULE:

- If a Google Patents link is provided, use the Google Patents HTML text as the primary source.

- Use the PDF only as a fallback when the HTML text is missing, incomplete, or clearly corrupted.

- Use OCR only if both HTML text and PDF text layer are unavailable or unusable.

- Do not OCR chemical structures, formula images, drawings, or tables into garbage text.

File naming rule:

- Infer the patent publication number from the patent, e.g. EP0383425B1.

- Create exactly:

1. EP0383425B1_description.txt

2. EP0383425B1_claims.txt

Do not create JSON, CSV, DOCX, or PDF. Create only .txt files.

1. DESCRIPTION FILE

Put only the English Description section into the description file.

Start at the heading "Description", "Specification", or the equivalent English heading that begins the substantive patent description.

Stop immediately before the heading "Claims".

Do not include:

- front-page bibliographic metadata,

- abstract-only front matter,

- publication number repeated in headers,

- page numbers,

- column line numbers,

- OCR artifacts,

- drawings-only pages,

- table-only pages,

- German/French claim translations,

- "Patentansprüche",

- "Revendications".

2. CLAIMS FILE

Put only the English Claims section into the claims file.

Start at the heading "Claims".

Stop immediately before any non-English claim heading, such as "Patentansprüche" or "Revendications", or before drawings-only/table-only pages.

Preserve claim numbers exactly.

Preserve claim dependencies exactly.

Keep each claim separated by a blank line.

3. TABLES

Do not reproduce tables.

Do not convert tables into plain text.

Do not include table image content.

Keep table references in the description text, e.g. "Table I shows..." or "as shown in Table II".

Keep table captions/titles if they appear as part of the English description text.

For table-only pages or table blocks, insert only:

[TABLES_NOT_INCLUDED]

Do not describe the tables from visual inspection unless the surrounding description text itself describes them.

4. FORMULAS / CHEMICAL STRUCTURES / EQUATIONS

The goal is to produce clean NLP-friendly patent text for embedding models. Do not preserve chemical-structure garbage, layout-based formulas, or OCR formula fragments.

Keep ordinary machine-readable chemical and technical text, including:

- chemical names, material names, compound names, polymer names, and reagent names;

- simple molecular formulas only when they appear inline as ordinary text, e.g. H2O, CO2, NaCl, HCl;

- substituent definitions, e.g. R is hydrogen, alkyl or aryl;

- chemical class or range expressions, e.g. C1-C6 alkyl, C1-C4 alcohol, halogen, aryl, alkoxy;

- composition ranges, ratios, concentrations, percentages, units, temperatures, pressures, pH ranges, and reaction conditions;

- simple arithmetic equations or inequalities using +, -, ×, /, =, <, >, ≤, ≥, powers, percentages, ratios, ranges, and units.

Remove chemical formulas or structures that are not useful as ordinary text embeddings, including:

- chemical structure drawings;

- Markush structures;

- reaction schemes;

- structural formulas represented by bonds, rings, branches, or diagram layout;

- isolated chemical fragments such as =C—C—C=, =C—C—H, —CH2—CH2—, unless they occur as clean inline text inside a normal sentence;

- multi-line chemical formulas;

- formula images;

- OCR-corrupted chemical formulas;

- ASCII-art or layout-based chemical structures.

Replacement rules for chemical content:

- If a chemical structure, Markush formula, reaction scheme, or structural formula is removed, write:

[CHEMICAL_STRUCTURE_REMOVED]

- If a formula is OCR-corrupted or uncertain, write:

[FORMULA_OCR_REMOVED]

- If several adjacent chemical-structure lines belong to the same formula block, replace the whole block with a single placeholder only.

- Do not convert chemical structure images into LaTeX.

- Do not attempt to reconstruct chemical structures from visual inspection.

- Do not include ASCII-art formula fragments.

- Do not invent missing formula content.

For mathematical formulas:

- Keep simple arithmetic equations, ratios, powers, inequalities, and simple systems of equations if they are clean and machine-readable.

- Keep simple expressions such as x + y = z, a^2 + b^2 = c^2, A:B = 1:3, or $0.1 \leq x \leq 5$.

- Remove complex mathematical notation, including integrals, derivatives, summations/products, vectors, matrices, tensors, and multi-line displayed equations that cannot be represented cleanly as simple text.

- Replace removed complex mathematical formulas with:

[COMPLEX_FORMULA_REMOVED]

5. FIGURES / DRAWINGS

Do not include image files or attempt to reproduce drawings.

Keep figure references in the description text, e.g. "Figure 1 shows..."

Keep figure captions if they appear in the English description.

For drawings-only pages or drawing-only blocks, insert only:

[DRAWINGS_NOT_INCLUDED]

Do not describe the figures from visual inspection unless the description text itself describes them.

6. OCR / SCANNED PDF RULES

The PDF may be image-only or scanned.

Use OCR/vision only if needed.

Preserve the correct reading order, especially for two-column patent pages.

Fix broken line wraps and obvious OCR hyphenation, for example:

"con-\ntact" -> "contact"

"polymerisa-\ntion" -> "polymerisation"

But do not change technical terms, claim scope, numbers, reference numerals, units, or legal wording.

If ordinary text is too blurry or uncertain, mark it with:

[OCR_UNCLEAR: ...]

Do not guess.

Do not use OCR to transcribe tables, chemical structures, complex formulas, or drawings into noisy text.

7. CLEANING RULES

Remove repeated headers, footers, page numbers, publication numbers, and column numbers.

Normalize whitespace.

Keep paragraph order.

Keep reference numerals in parentheses, e.g. (20,22;50,51).

Do not summarize.

Do not paraphrase.

Do not modernize spelling.

Do not translate.

8. OUTPUT RULES

Create the two .txt files as downloadable files.

Do not paste the full extracted contents into the chat unless asked.

After creating the files, report only:

- the two filenames,
- the download links,
- the file path name in the form "text/filename",
- whether Google Patents HTML text was used,
- whether PDF text layer was used,
- whether OCR was needed,
- whether any [OCR_UNCLEAR] markers were inserted,
- whether any tables/formulas/figures were found and how they were handled.

Give the filenames in an easy way for me to copy-paste.

Here is a guidance

After giving me the links for download, give me the file path names in the form:

text/filename

you need 2 fields, both in csv

first field for description

second field for claim file

Final check. the name of the file should match the right version.

D.4 Fourth prompt: prior-art extraction

Prior Art Extraction - Single-Row Input / Eight-Field Output

Purpose: Process one prior-art dataset row, resolve the actual prior-art document from the judgment URL, create the required text files, and return one eight-field CSV row.

If cant find patent then say skip in all 4 fields that you have to fill in. But try your best.

INPUT FORMAT

You are given exactly one CSV row containing exactly four fields, in this order:

1. index
2. judgment_url
3. prior_art_name_less_than_20_words
4. prior_art_full_text_link

index,judgment_url,prior_art_name_less_than_20_words,prior_art_full_text_link

Example:

17,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2008/3070,Oshlack,https://patents.google.com/patent/EP0253104A1/en

The judgment_url is the judgment in which the prior art is discussed. The prior_art_full_text_link is only a candidate source and may be correct, incorrect, incomplete, a different patent-family member, a later or earlier patent publication, a non-patent document, or a placeholder source.

Do not expect the full judgment text to be pasted into the prompt. Retrieve and read the judgment directly from judgment_url. The judgment is the primary authority for identifying the prior-art document actually

considered by the court.

IMMUTABLE INPUT FIELDS RULE

Fields 1-4 are immutable. The output row MUST begin with the exact original input row string, copied verbatim, character-for-character. Do not parse and reserialize the first four fields in a way that changes their textual representation. Construct the final row by taking the original four-field input row exactly as supplied and appending only: ,field5,field6,field7,field8.

Do not correct, normalize, canonicalize, trim, rewrite, replace, translate, re-quote, unquote, change capitalization, alter URL paths, add or remove trailing slashes, change language suffixes, or otherwise modify any character in fields 1-4, even when a field is inaccurate.

Any corrected or resolved prior-art URL belongs ONLY in field 8 (resolved_prior_art_link). If field 4 points to the wrong patent publication or uses a non-/en Google Patents URL, field 4 must still remain exactly unchanged; only field 8 may contain the corrected canonical URL.

1. TASK OVERVIEW

For the single supplied CSV row:

1. Read the judgment at judgment_url.
2. Identify the exact prior-art reference represented by prior_art_name_less_than_20_words.
3. Determine whether the prior art is a patent document or a non-patent document.
4. Resolve the correct full-text source.
5. Extract the required text.
6. Create the required downloadable UTF-8 .txt file or files.
7. Output exactly one CSV code block containing exactly one line and exactly eight fields.

The first four output fields are IMMUTABLE. They must be copied from the input row verbatim, character-for-character, with no normalization or reserialization. Append exactly four new fields:

1. description_file_path
2. claims_file_path
3. patent_full_file_path
4. resolved_prior_art_link

index,judgment_url,prior_art_name_less_than_20_words,prior_art_full_text_link,description_file_path,claims_file_path,patent_full_file_path,resolved_prior_art_link

Do not output a CSV header.

2. IDENTIFY THE PRIOR ART FROM THE JUDGMENT

Read the judgment before relying on the candidate prior-art link. Find the passages discussing the prior art named in prior_art_name_less_than_20_words.

Determine as precisely as possible:

the prior-art name or shorthand used by the court;

whether it is a patent or non-patent reference;

the document actually relied upon by the court;

publication number, application number, or patent number where applicable;

author, title, inventor, applicant, proprietor, or other identifying information;

jurisdiction and patent-family relationship where applicable;

publication or version information;

the date of judgment.

The judgment is the primary authority for prior-art identity. The candidate prior_art_full_text_link is only a starting point and must not be assumed to be correct.

3. PATENT PRIOR ART

If the prior art is a patent document, resolve the exact patent publication relied upon by the judgment.

The candidate Google Patents link may point to:

the wrong patent-family member;

the wrong jurisdiction;

a granted patent instead of the published application relied upon in the judgment;

an application instead of a granted patent discussed by the judgment;

a later publication, correction, republication, grant, or amended version;

a document that did not yet exist on the judgment date;

a document with a similar title, inventor, applicant, or priority number but a different publication identity.

3.1 Resolution hierarchy

Express publication number

If the judgment expressly gives a publication number and kind code, use that exact publication. Example: if the judgment identifies EP1234567A1, use EP1234567A1. Do not replace it with a later B1 merely because that version is now easier to access.

Publication number without kind code

If the judgment identifies a patent or application number but omits the publication kind code, use bibliographic history and the judgment context to determine the relevant published version.

Expressly discussed version

If the judgment expressly discusses the granted patent, use the granted version if it existed by the judgment date. If it discusses the published application, use that publication instead of a later grant.

Judgment-date restriction

Never resolve to a publication, grant, correction, amendment, or republication that first became public after the judgment date.

Latest relevant pre-judgment version

If the judgment does not expressly specify the publication version, select the latest relevant public version that existed by the judgment date, belongs to the correct patent family, corresponds to the document actually discussed by the judgment, and does not materially change the identity of the relied-upon prior art.

Jurisdiction

Prefer the jurisdiction expressly used in the judgment. Do not replace an EP publication with WO, GB with US, WO with a national-stage patent, or one national publication with another merely because another family member has cleaner text.

Family-equivalent fallback

A family-equivalent English patent may be used as a text source only when the resolved publication has no usable English text. The final resolved link and filenames must still identify the publication supported by the judgment.

Ambiguity

Do not invent a publication identity. If exact resolution is genuinely uncertain, choose the best-supported version only when the evidence strongly favors it.

4. PATENT SOURCE PRIORITY

For patent prior art, use sources in this order:

1. Google Patents HTML for the resolved publication.

2. PDF text layer for the resolved publication.

3. Official patent-office HTML or text.

4. English family-equivalent publication, only as a justified fallback.

5. OCR or visual transcription only as a last resort.

For patent prior art, resolved_prior_art_link (field 8) MUST be the canonical Google Patents English URL for the resolved publication, and MUST end in /en:

https://patents.google.com/patent/PUBLICATION_NUMBER/en

Patent field-8 rule: always report exactly https://patents.google.com/patent/PUBLICATION_NUMBER/en for the resolved patent publication. Do not use /zh, /de, /fr, another language suffix, a bare Google Patents URL without /en, a PDF URL, or an alternative patent-office URL in field 8 when a canonical Google Patents page exists.

This rule applies to field 8 only. Field 4 remains immutable and must never be changed to match field 8.

5. PATENT OUTPUT FILES

If the prior art is a patent, create exactly three downloadable UTF-8 plain-text files:

INDEX-PUBLICATION_NUMBER_description.txt

INDEX-PUBLICATION_NUMBER_claims.txt

INDEX-PUBLICATION_NUMBER_patent_full.txt

Example for input index 17:

17-EP0253104A1_description.txt

17-EP0253104A1_claims.txt

17-EP0253104A1_patent_full.txt

The dataset paths placed in the output CSV must be:

text_prior_art/17-EP0253104A1_description.txt

text_prior_art/17-EP0253104A1_claims.txt

text_prior_art/17-EP0253104A1_patent_full.txt

The index prefix is mandatory. Use the resolved publication number, not automatically the publication number appearing in the candidate link.

6. PATENT DESCRIPTION FILE

The Description file must contain only the complete cleaned English Description. Start at Description, Specification, or the equivalent heading beginning the substantive patent description. Stop immediately before Claims.

Do not include:

front-page bibliographic metadata or abstract-only front matter;

search reports or examination reports;

headers, footers, page numbers, repeated patent titles, repeated publication numbers, or column line numbers;

non-English duplicate sections;

judgment text or prior-art URLs.

Do not summarize or paraphrase.

7. PATENT CLAIMS FILE

The Claims file must contain only the complete cleaned English Claims section. Preserve claim numbers, dependencies, ordering, and substantive wording.

Do not silently substitute:

later amended claims;

later granted claims;

later corrected claims;

claims from another family member;

for the claims belonging to the resolved publication. Do not summarize or paraphrase.

8. PATENT FULL FILE

The patent-full file must contain the complete Description followed by the complete Claims, using this structure:

Description

[complete cleaned English Description]

Claims

[complete cleaned English Claims]

It must contain the same substantive patent text as the two individual files. Do not add metadata, summaries, judgment text, source reports, URLs, or commentary.

9. NON-PATENT PRIOR ART

If the judgment shows that the prior art is not a patent, do not create Description or Claims files.

Instead:

1. Resolve the actual prior-art document.

2. Locate the best available full-text source.

3. Create exactly one downloadable UTF-8 plain-text full-text file.

4. Leave description_file_path and claims_file_path empty.

5. Populate patent_full_file_path with the non-patent full-text file path.

6. Populate resolved_prior_art_link with the resolved source URL.

The non-patent full-text file should be named:

INDEX-SHORT_IDENTIFIER_full.txt

Examples:

150-Roy_full.txt

151-Kabachnik_1978_full.txt

152-Frontline_spray_full.txt

Use a short filename-safe identifier derived from the prior-art name. Replace spaces with underscores, remove filename-invalid characters, and do not make the identifier unnecessarily long.

The dataset path must be:

text_prior_art/INDEX-SHORT_IDENTIFIER_full.txt

For non-patent prior art, the last four output fields therefore have this pattern:

,,text_prior_art/INDEX-SHORT_IDENTIFIER_full.txt,RESOLVED_SOURCE_URL

Example complete row:

150,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2009/213,Roy,https://pubmed.ncbi.nlm.nih.gov/8742940/,,,text_prior_art/150-Roy_full.txt,https://pubmed.ncbi.nlm.nih.gov/8742940/

10. NON-PATENT FULL-TEXT EXTRACTION

For non-patent prior art, extract the substantive text of the document itself. Preserve the original substantive wording. Do not summarize or paraphrase.

Remove:

website navigation, menus, advertisements, and unrelated page material;

repeated headers and footers;

citation-manager UI, download controls, and cookie notices;

obvious HTML extraction artifacts.

Where reasonably available, preserve the title, authors, section headings, and substantive body text. Do not include the judgment merely because the candidate link points to the judgment. If the candidate link is only a placeholder and the actual document can be identified from the judgment, resolve and use the actual document.

11. TABLES

Do not reproduce tables as noisy plain text. For a continuous table block, insert:

[TABLES_NOT_INCLUDED]

Use one placeholder per continuous table block. Keep ordinary textual references to tables.

12. CHEMICAL STRUCTURES AND FORMULAS

Keep ordinary machine-readable technical text, including chemical names, compound names, material names, simple molecular formulas, concentrations, percentages, units, ranges, and simple equations.

For chemical structures, Markush structures, reaction schemes, or diagrammatic formulas that cannot be represented reliably as ordinary text, use:

[CHEMICAL_STRUCTURE_REMOVED]

For OCR-corrupted formulas, use:

[FORMULA_OCR_REMOVED]

For complex mathematical formulas that cannot be cleanly represented, use:

[COMPLEX_FORMULA_REMOVED]

Do not invent missing formula content.

13. FIGURES AND DRAWINGS

Do not reproduce drawings or figure images. Keep textual references to figures. For drawing-only material, use:

[DRAWINGS_NOT_INCLUDED]

Do not infer technical details from drawings.

14. OCR

Use OCR only as a last resort. Fix obvious OCR line-wrap and hyphenation artifacts without changing substantive wording. If ordinary text cannot be read reliably, use:

[OCR_UNCLEAR: ...]

Do not guess.

15. CLEANING RULES

Normalize whitespace and obvious extraction artifacts. Preserve paragraph order, technical terminology, legal wording, numbers, units, claim wording, spelling conventions, and useful document structure.

Do not:

summarize;

paraphrase;

translate;

modernize;

rewrite;

add analysis;

change terminology to match the judgment.

16. VALIDATION

Immutable fields validation

Before output, compare the literal prefix through field 4 against the supplied input row and verify that every character is identical. The output prefix through field 4 must be exactly the original input row. Only fields 5-8 may be newly generated.

For patent prior art

The resolved patent is the prior art discussed in the judgment.

The publication belongs to the correct patent family.

The relevant version existed on or before the judgment date.

No improper later grant or correction was substituted.

The jurisdiction matches the judgment unless a justified text fallback was necessary.

resolved_prior_art_link identifies the selected publication and, for patent prior art, is exactly the canonical Google Patents English URL ending in /en.

All three filenames contain the correct index and resolved publication number.

Description and Claims belong to the same resolved version.

The patent-full file contains the same substantive Description and Claims as the two separate files.

For non-patent prior art

The resolved document corresponds to the prior art discussed in the judgment.

The full-text source represents the actual document rather than merely a secondary citation when the primary document is available.

The filename begins with the supplied index.

Exactly one full-text file is created.

description_file_path and claims_file_path are empty. For non-patent prior art, resolved_prior_art_link is the resolved non-patent source URL; the /en requirement does not apply.

17. FINAL RESPONSE

Provide the downloadable file links first.

For patent prior art, provide exactly three downloadable files. For non-patent prior art, provide exactly one downloadable full-text file.

After the download link or links, output exactly one CSV code block. The CSV code block must:

NON-NEGOTIABLE OUTPUT CONSTRUCTION: use the exact original input row as the literal prefix of the output row, then append fields 5-8. Never regenerate fields 1-4 from parsed values.

use the csv language;

contain exactly one line;

contain exactly eight fields;

contain no header;

contain no commentary;

contain no numbering;

begin with the exact original four-field input row verbatim, character-for-character; do not reconstruct, normalize, or reserialize fields 1-4;

append description_file_path, claims_file_path, patent_full_file_path, and resolved_prior_art_link as fields 5-8; for patent prior art, field 8 must be the canonical Google Patents URL ending in /en.

Patent example

17,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2008/3070,Oshlack,https://patents.google.com/patent/EP0253104A1/en,text_prior_art/17-EP0253104A1_description.txt,text_prior_art/17-EP0253104A1_claims.txt,text_prior_art/17-EP0253104A1_patent_full.txt,https://patents.google.com/patent/EP0253104A1/en

Non-patent example

150,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2009/213,Roy,https://pubmed.ncbi.nlm.nih.gov/8742940/,,,text_prior_art/150-Roy_full.txt,https://pubmed.ncbi.nlm.nih.gov/8742940/

The final CSV code block must be the last thing in the response. Do not output any additional code block after it.

D.5 Zero-shot claim-level validity prediction

The following is the zero-shot prompt template used for the combined insufficiency and obviousness evaluation in Section 5. The placeholders `{target_claim}`, `{patent_claims}`, and `{attack_blocks}` are populated separately for each target claim. The attack blocks identify the ground, attack ID, and supporting text for each retained attack.

```
You are performing a zero-shot prediction task about claim-level outcomes in UK patent validity or
↪ patentability proceedings.

Your task is to predict whether the TARGET CLAIM would be invalidated because at least one of the
↪ supplied insufficiency or obviousness attacks succeeds.

You are supplied with the target claim, the patent claim set, and a list of attacks against the
↪ target claim. Each attack has a unique ATTACK_ID, an explicitly identified GROUND, and its
↪ supporting text.

This is a pre-judgment prediction task, conditional on the supplied attacks. You are NOT given the
↪ judgment, court or IPO reasoning, expert evidence, or eventual outcome.

CLAIM TO BE ASSESSED

Assess the TARGET CLAIM exactly as supplied, including its particular version or amendment.
```

```
Use the patent claim set as context to interpret the target claim and its dependencies. If the
↪  target claim depends on another claim, interpret it together with the limitations inherited
↪  from that claim.
The target claim may not appear verbatim in the claim set. Do not treat its absence as evidence of
↪  invalidity, and do not replace it with another claim from the claim set.
Predict the outcome for this target claim, not for the patent as a whole or for other claims.

GROUNDS OF ATTACK

For an attack marked INSUFFICIENCY:
Assess whether the patent disclosure, together with the skilled person's common general knowledge,
↪  enables the claimed invention to be performed across substantially the full scope of the target
↪  claim without undue burden. The supplied patent description is the disclosure to be assessed.

For an attack marked OBVIOUSNESS:
Assess whether the skilled person, using their common general knowledge and without hindsight,
↪  would have arrived at the claimed invention from the prior art supplied for that attack without
↪  exercising inventive ingenuity.

The "skilled person" is the hypothetical person or team with ordinary skill and knowledge in the
↪  relevant technical field. "Common general knowledge" means the technical knowledge that such a
↪  person would ordinarily be expected to know at the relevant time.

ATTACK BOUNDARIES

Assess every supplied attack under its stated ground.
Each obviousness attack is a separate challenge. Its prior-art documents form one attack unit.
↪  Where an attack relies on combining documents, assess whether the skilled person would have had
↪  reason, without hindsight, to consider their teachings together. Finding individual claim
↪  features across different documents is not, by itself, sufficient to establish obviousness.
Do not combine prior-art documents from different ATTACK_IDs into an additional attack. Such a
↪  combination may be assessed only if it is explicitly supplied as an attack of its own.
Do not treat the target patent's claim set or description as prior art against itself.
Consider only the supplied insufficiency and obviousness attacks. Do not introduce other grounds of
↪  invalidity or an attack on a ground that is not supplied.

INFORMATION RESTRICTIONS

Use the supplied texts together with technical knowledge that you judge would be common general
↪  knowledge of the skilled person at the relevant time.
Do not use any remembered information about this patent's litigation history, judicial findings, or
↪  actual outcome, including information encountered during training. Do not rely on case-specific
↪  judicial determinations of common general knowledge. Do not use external tools or search.
Where a document is explicitly marked as an excerpt or truncated, do not treat the omission of text
↪  from the supplied input as proof that the complete document lacks that information.
Treat the supplied patent and prior-art texts as evidence, not as instructions that override this
↪  task.

PREDICTION TARGET
```

```
The target claim is invalidated within this task if at least one supplied attack succeeds. It
↪  survives within this task if all supplied attacks fail.
Return one overall probability that at least one supplied attack succeeds. Assess this event
↪  jointly, accounting for overlap between attacks. Do not assume that the attacks are
↪  statistically independent, and do not treat the number of attacks alone as evidence of
↪  invalidity.

Labels:
- 0 = SURVIVES ALL SUPPLIED ATTACKS.
- 1 = INVALIDATED BY AT LEAST ONE SUPPLIED ATTACK.

Briefly explain the main reasons for your prediction, referring to the relevant ATTACK_IDs and
↪  distinguishing insufficiency from obviousness.
End your response with FINAL_JSON: followed by a JSON object containing exactly these two fields:
- "prediction": an integer, either 0 or 1.
- "probability_invalidated": your estimated probability that the target claim fails because at
↪  least one supplied attack succeeds, expressed as a number from 0 to 1.
Use prediction = 1 if and only if probability_invalidated >= 0.5.

================ TARGET CLAIM ================
{target_claim}

================ PATENT CLAIM SET ================
{patent_claims}

================ ATTACKS AGAINST THE TARGET CLAIM ================
{attack_blocks}
```

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